



SupraHD®-748-DTS-GEN-v1.1

SupraHD[®]

ZR39748

Hardware Datasheet

Document Version 1.1

8/6/2010

**DO NOT COPY
CONFIDENTIAL**

The information contained in this manual is Preliminary in nature and while every effort has been made to provide complete and accurate data, there may be some areas of the manual that contain incomplete information. In addition, due to the current preliminary stage of the product the information in this manual may change.



SupraHD®-748-DTS-GEN-v1.1

COPYRIGHT AND LICENSING REQUIREMENTS

© 2003 - 2010 Zoran Corporation. All rights reserved. Zoran and the Zoran logo are trademarks of Zoran Corporation. All other brand product names and company names are trademarks of their respective owners.

This datasheet contains information that is confidential to Zoran Corporation, and is subject to the terms and conditions, including confidentiality obligations, set forth in the applicable Nondisclosure Agreement between the User and Zoran Corporation and or its subsidiaries.

Information herein is subject to change without notice. Zoran Corporation does not assume any responsibility for any use of, or reliance on, the information contained herein.

THIS MANUAL AND ALL INFORMATION CONTAINED HEREIN IS PROVIDED "AS IS" WITHOUT WARRANTY OF ANY KIND, WHETHER EXPRESS, IMPLIED, STATUTORY, OR OTHERWISE. ZORAN CORPORATION SPECIFICALLY DISCLAIMS ALL IMPLIED WARRANTIES OF MERCHANTABILITY, NONINFRINGEMENT, AND FITNESS FOR A PARTICULAR PURPOSE.

Features and capabilities defined in this document are enabled and controlled by a combination of hardware in the device and software control. Some of these features and capabilities may NOT be supported by the current software available from Zoran Corporation. Please contact your Zoran representative for details.

Zoran Corporation
1390 Kifer Road
Sunnyvale, CA 94086-5305
Tel: +1 (408) 523-6500
Fax: +1 (408) 523-6501
<http://www.zoran.com/products/dtv/>

Document ID: SupraHD-748-DTS-GEN-v1.1



SupraHD®-748-DTS-GEN-v1.1

Dolby®

Dolby and the double-D symbol are trademarks of Dolby Laboratories.

Supply of this Implementation of Dolby Technology does not convey a license nor imply a right under any patent, or any other Industrial or Intellectual Property Right of Dolby Laboratories, to use this Implementation in any finished end-user or ready-to-use final product. It is hereby notified that a license for such use is required from Dolby Laboratories.

Companies planning to use this implementation in products must obtain a license from Dolby Laboratories Licensing Corporation before designing such products. Additional per-chip royalties may be required and are to be paid by the purchaser to Dolby Laboratories, Inc. Details of the OEM Dolby Digital license may be obtained by writing to:

Dolby Laboratories, Inc.

Dolby Laboratories Licensing Corporation

Attn: Intellectual Property Manager

100 Potrero Avenue

San Francisco, CA 94103-4813

TEL: +1 (415) 558-0200

FAX: +1 (415) 863-1373

<http://www.dolby.com>



SupraHD®-748-DTS-GEN-v1.1

MPEG LA, L.L.C. Notice

USE OF THIS PRODUCT IN ANY MANNER THAT COMPLIES WITH THE MPEG-2 STANDARD IS EXPRESSLY PROHIBITED WITHOUT A LICENSE UNDER APPLICABLE PATENTS IN THE MPEG-2 PATENT PORTFOLIO, WHICH LICENSE IS AVAILABLE FROM MPEG LA, L.L.C., 250 STEELE STREET, SUITE 300, DENVER, COLORADO 80206.



Table of Contents

Chapter 1. SupraHD® 748 Overview	2
1.1. SupraHD® 748 IC Description	2
1.2. SupraHD® 748 Features.....	2
1.2.1. Embedded Processing Unit.....	2
1.2.2. Video Processing	3
1.2.3. Video Input	4
1.2.4. Video Output	5
1.2.5. Audio Processing and I/O	6
1.2.6. Video Decoders.....	7
1.2.7. Front-End Demod / Demux	8
1.2.8. Memory Support	9
1.2.9. Integrated TV MicroController.....	9
1.2.10. Graphics Processing	9
1.2.11. System Interfaces	10
1.2.12. Security Features	10
1.2.13. Misc. IC Information	10
1.3. SupraHD® 748 Block Diagrams.....	11
Chapter 2. Signal Listings	14
2.1. BGA Package Signal Lists	14
2.1.1. Signal Listing by Ball Order.....	14
2.1.2. Signals with Alternate Functions	24
2.1.3. Signal Listings by Function	26
2.2. QFP Package Signal Lists.....	42
2.2.1. Signal Listing by Pin Order.....	42
2.2.2. Signals with Alternate Functions	49
2.2.3. Signal Listings by Function	51
2.3. Differences Between BGA and QFP Packages	64
Chapter 3. Operating Conditions	66
3.1. Heat Characterization	66
3.2. Absolute Maximum Ratings.....	66
3.3. Power Dissipation.....	67
3.4. Recommended DC Operating Conditions	68
3.5. DDR2 Electrical Characteristics	69
3.6. Digital Video Electrical Characteristics.....	70



Chapter 4. SupraHD® 748 Signal Maps, Packaging and Ordering Information	71
4.1. SupraHD® 748 Order Information	71
4.2. SupraHD® 748 Ball and Pin Maps	71
4.3. BGA Package Mechanicals	75
4.4. QFP Package Mechanicals	79
Appendix A: Change History	82
Appendix B: Document Improvement.....	83



SupraHD®-748-DTS-GEN-v1.1

List of Figures

Figure 1:	Basic SupraHD® 748 IC Block Diagram.....	1
Figure 2:	SupraHD® 748 Typical ATSC TV System Implementation Block Diagram	11
Figure 3:	SupraHD® 748 Detailed Block Diagram	12
Figure 4:	SupraHD® 748 I/O Option Dataflows	13
Figure 5:	ZR39748 BGA Ballout - Top View, Looking Down (Part 1 of 2).....	72
Figure 6:	ZR39748 BGA Ballout - Top View, Looking Down (Part 2 of 2).....	73
Figure 7:	ZR39748 QFP Pins - Top View, Looking Down	74
Figure 8:	ZR39748 BGA Physical Package (Top View)	75
Figure 9:	ZR39748 BGA Physical Package (Bottom View).....	76
Figure 10:	ZR39748 BGA Physical Package (Side View)	77
Figure 11:	ZR39748 QFP Physical Package (Top & Side Views)	79
Figure 12:	ZR39748 QFP Physical Package (Section Views).....	80



List of Tables

Table 1:	SupraHD® 748 BGA Signals – Ball Order.....	14
Table 2:	SupraHD® 748 BGA Signals with Alternate Functions	24
Table 3:	SupraHD® 748 BGA Signals – JTAG/EJTAG	26
Table 4:	SupraHD® 748 BGA Signals – Analog Video Front-End Input.....	26
Table 5:	SupraHD® 748 BGA Signals – HDMI Input.....	28
Table 6:	SupraHD® 748 BGA Signals – LVDS & TCON Output.....	29
Table 7:	SupraHD® 748 BGA Signals – Audio Input/Output	31
Table 8:	SupraHD® 748 BGA Signals – USB	32
Table 9:	SupraHD® 748 BGA Signals – DDR2	33
Table 10:	SupraHD® 748 BGA Signals – SPI.....	34
Table 11:	SupraHD® 748 BGA Signals – UART & I2C	34
Table 12:	SupraHD® 748 BGA Signals – General Purpose I/O	35
Table 13:	SupraHD® 748 BGA Signals – TVuC and Main CPU Signals	36
Table 14:	SupraHD® 748 BGA Signals – Clocks.....	37
Table 15:	SupraHD® 748 BGA Signals – Power and Ground.....	38
Table 16:	SupraHD® 748 QFP Signals – Pin Order	42
Table 17:	SupraHD® 748 QFP Signals with Alternate Functions	49
Table 18:	SupraHD® 748 QFP Signals – JTAG/EJTAG	51
Table 19:	SupraHD® 748 QFP Signals – Analog Video Front-End Input.....	51
Table 20:	SupraHD® 748 QFP Signals – HDMI Input	52
Table 21:	SupraHD® 748 QFP Signals – LVDS & TCON Output.....	54
Table 22:	SupraHD® 748 QFP Signals – Audio Input/Output	55
Table 23:	SupraHD® 748 QFP Signals – USB	56
Table 24:	SupraHD® 748 QFP Signals – DDR2.....	57
Table 25:	SupraHD® 748 QFP Signals – SPI	58
Table 26:	SupraHD® 748 QFP Signals – UART & I2C	58



Table 27:	SupraHD [®] 748 QFP Signals – General Purpose I/O	59
Table 28:	SupraHD [®] 748 QFP Signals – TVuC and Main CPU Signals.....	60
Table 29:	SupraHD [®] 748 QFP Signals – Clocks.....	61
Table 30:	SupraHD [®] 748 QFP Signals – Power and Ground.....	61
Table 31:	SupraHD [®] 748 BGA Signals Absent from QFP Package	64
Table 32:	Absolute Maximum Ratings.....	66
Table 33:	Power Dissipation SupraHD [®] 748.....	67
Table 34:	Maximum Current Dissipation Per Supply: SupraHD [®] 748	67
Table 35:	Standby Power Dissipation SupraHD [®] 748.....	68
Table 36:	Recommended DC Operating Conditions	68
Table 37:	DDR2 Interface Absolute Maximum Ratings	69
Table 38:	DDR2 DC Electrical Specifications	70
Table 39:	DDR2 AC Electrical Specifications	70
Table 40:	Primary Digital Video I/O Electrical Specifications	70
Table 41:	BGA Package Physical Parameters	78
Table 42:	QFP Package Physical Parameters	81
Table 43:	History of Document Changes	82

Zoran ZR39748

- **Integrated Digital & Analog Demodulator**

- 8VSB/QAM-B
- NTSC/BTSC/A2K

- **Video Inputs**

- Three (3) 1080p HDMI (v1.4a/DC)*
- One (1) 1080p YPbPr
- One (1) VGA, up to WUXGA resolution
- Two (2) CVBS*, One (1) S-Video

- **Audio Inputs**

- Five (5) stereo L/R line-level*

- **Internal Video/Audio Processing**

- NTSC/PAL/SECAM video decoder
- MPEG-2 decoder
- 10-bit video processing
- 1080i motion-adaptive de-interlacer
- ACM-2D color processor
- Graphics blending/overlay
- Audio DSP

- **Video Outputs**

- Dual-channel LVDS (1080p, up to 10bpp)
- miniLVDS & RSDS (6/8bpp, up to 330MHz)
- LCD panel timing control signals (TCON)

- **Audio Outputs**

- One (1) stereo L/R DDX differential
- One (1) stereo L/R single-ended DDX
- Optional up-to-four (4) more single-ended DDX
- Optional up-to-three (3) I2S stereo pairs
- One (1) S/PDIF

- **System Processors & Interfaces**

- 300MHz system CPU
- TV microcontroller for standby mode
- One (1) USB 2.0

- **External SPI Flash Memory: 2-16MByte**

- 2-4MB typical for ATSC DTV application

- **External 16-bit DDR2 Required**

- DDR2-800 for most design configurations
- DDR2-1066 for 1080p with overdrive designs
- 64MByte typical for most designs

- **Power**

- 1.1V core voltage, 1.8V memory I/F, 3.3V I/O

- **Two Package Options**

- 365-ball BGA, 23x23mm²
- 256-pin LQFP with e-pad, 28x28mm²

(*) Slight variation of support with QFP package

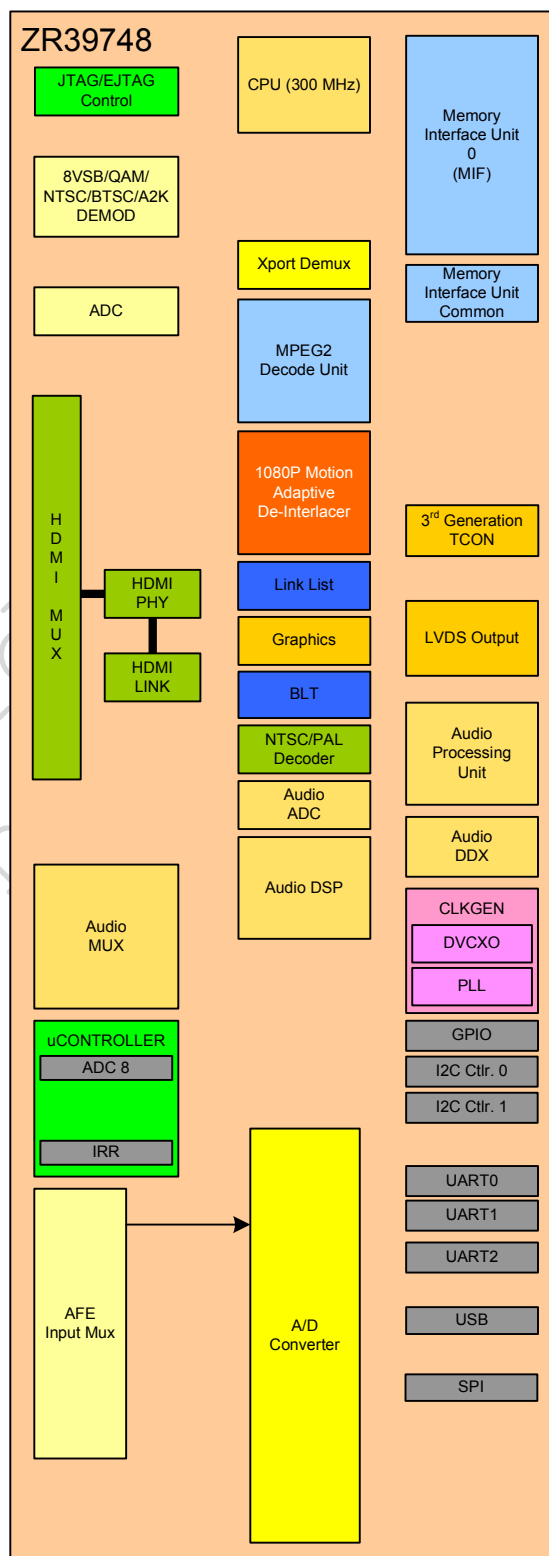


Figure 1: Basic SupraHD® 748 IC Block Diagram

Chapter 1. SupraHD® 748 Overview

1.1. SupraHD® 748 IC Description

The SupraHD® 748 is a member of the SupraHD® family of DTV system-on-chip (SoC) developed by Zoran. This device is intended to be used in ATSC high-definition digital television implementations. This device includes all of the functionality required to support the television implementations shown in the following block diagrams.

Figure 2 shows a typical ATSC system implementation using the SupraHD® 748.

Figure 3 shows the detailed block diagram of the SupraHD® 748.

Figure 4 shows the video and audio input/output options of the SupraHD® 748.

1.2. SupraHD® 748 Features

The following sub-sections list the features of the SupraHD® 748 per category. Note that features unique to the BGA package are indicated with a "(BGA package)" designation while QFP package features are indicated with "(QFP package)".

1.2.1. Embedded Processing Unit

- **High performance CPU**
 - Integrated high-performance MIPS® 4KEc™ CPU operating at 300MHz
 - 32-bit MIPS32 enhanced architecture
 - 8 K instruction cache, 8 K data cache, (2-way set associative)
 - Programmable memory management unit
 - Multiply/Divide unit
- **Power-down mode (triggered by WAIT instruction)**
- **EJTAG debug support**
- **Fully production-tested software suite**
 - ATSC/NTSC DTV application with customizable OSD
 - V-Chip for analog and digital channels
 - PSIP parsing for channel map and EPG
 - Analog and digital closed-captioning (EIA-608 and EIA-708)
 - Royalty-free Zoran True Fonts for OSD and closed-captioning
 - Transport, video decode (single MP@HL), audio decode (AC-3, MPEG Layer I & II), graphics, and display drivers
 - Drivers for tuner, HDMI and analog inputs
 - ThreadX royalty-free RTOS

1.2.2. Video Processing

- **Image processing**
 - Up to 10-bit processing
 - De-interlacing
 - 1080i capable, per-pixel motion adaptive, multiple cadence detection, 8° low-angle interpolation
 - Black bar detection
 - Horizontal and vertical
- **Image quality enhancements**
 - Noise reduction (up to 1080p)
 - Temporal
 - Spatial
 - Impulse
 - MPEG post-processing
 - De-blocking
 - Adaptive contrast control (histogram-based, fully-programmable)
 - Advanced Color Management 2D
 - Horizontal luma peaking with coring
 - Sharpness control
 - Vertical and horizontal LTI
 - Horizontal CTI
 - Y/C vertical peaking with adaptive coring
- **Video scaling and composition**
 - Horizontal scaler
 - 17-tap FIR, 64-phase FIR
 - Programmable up scaler [64x]
 - Waterglass scaler
 - Programmable down scaler [1/32x]
 - Non-linear scaler - 3-segment parabola, 17-tap FIR, 64-phase FIR
 - Letterbox support
 - Pan and Scan support
 - 10-bit processing
 - Vertical scaler
 - 5-tap FIR, 64-phase FIR
 - Programmable up scaler [64x]
 - Programmable down scaler [1/32x]

1.2.3. Video Input

- **Integrated HDMI link and PHY**
 - Three physical ports (BGA package)
 - One physical port (QFP package)
 - Single instance of the PHY
 - HDMI v1.4a-compliant
 - Supports up to 1080p input resolution
 - Standby power CEC monitor
 - Supports all DTV resolutions (480i / 576i / 480p / 576p / 720p / 1080i / 1080p)
 - Capable of carrying IEC61937 compressed audio (Dolby Digital, etc.)
 - Integrated High-bandwidth Digital Content Protection (HDCP) cipher
 - Direct capture of video, audio, and control information in distinct memory buffers
- **Integrated high definition (HD) capture/video inputs**
 - Color space conversion
 - Downscaling to either 4:2:2 or 4:4:4 output to memory
 - One (1) YPbPr input
 - Up to 165MHz sample rate (Up to 1080p)
 - Sync Modes: sync on green (SOG) or luma (SOY) input, mid-point and sync tip clamping
 - SOG or SOY inputs:
 - AC coupled
 - Low pass filter (500 KHz)
 - Dynamic range 0.5-2.0V
 - >1MΩ DC input impedance
 - Coast input and support
 - Activity / polarity detectors with timing measurement
 - HSYNC present
 - VSNC present
 - SOG/SOY present
 - 2nd YPbPr input available using S-Video and SIF lines (BGA package)
 - One (1) RGB input
 - Separate HSYNC, VSNC inputs
 - TTL level-compatible
 - Up to WUXGA (1920x1200x60Hz with reduced blanking)
 - Support for 10-bit processing
 - Up to 165 MHz input bandwidth
- **Standard definition (SD) video inputs**
 - Two (2) CVBS inputs (BGA package)
 - One (1) CVBS input (QFP package)
 - One (1) S-Video input
 - No low-pass filter (LPF) required on SD inputs

1.2.4. Video Output

- **Gradient recovery**
 - Up to 10-bit output for 8-bit video input
- **Overdrive**
 - Improves LCD response time
 - Proprietary Zoran scheme for applying overshoot / undershoot pixel values
- **Display processor**
 - Main output display formats include 1920x1080p, 1680x1050p, 1440x900p, 1366x768p, 1280x768p, 1280x720p and 1024x768p
 - Panel frame rate up to 60Hz support for 1920x1080 panel resolution
 - Output can support 6, 8 or 10-bit panels
 - EIA-608 and EIA-708 closed caption support
 - Horizontal and vertical flip support
- **Integrated dual-channel LVDS output for direct panel display support**
 - Supports up to 165MHz (see below for miniLVDS speed)
 - 1080p output flat panel support
 - 6, 8 and 10-bit panel support
 - Programmable PWM backlight control
 - Spread spectrum clock generation
 - $\pm 6.25\%$ clock modulation
- **Integrated Timing Controller (TCON) for direct panel timing control**
 - Up to 11 user-programmable timing control signals to drive source and gate drivers
 - Fail-safe circuit to protect panel from off-spec timing
 - miniLVDS dual-channel output with TCON signals activated
 - 330MHz single-channel miniLVDS support with TCON signals activated
 - RSDS single-channel output with TCON signals activated (BGA package)

1.2.5. Audio Processing and I/O

- **Five (5) L/R line-level stereo inputs**
 - Multiplexed into a single stereo ADC
 - 16-bit A/D conversion
 - 82dB dynamic range and -75dB THD A/D conversion
 - Supported audio sampling rates from 32 to 96 KHz
- **Up to six (6) channels of audio output, on DDX or I2S lines**
 - Two (2) DDX differential speaker outputs for direct power-stage drive (channels 0-1)
 - Or four (4) single-ended DDX for analog output (channels 0-3)
 - Or one (1) stereo I2S output (channels 0-1)
 - I2S data aligned in I2S format; Contact Zoran for left-justified format support
 - Two (2) single-ended DDX for line-out (channels 2-3)
 - Or two (2) single-ended DDX for analog output (channels 4-5 – only when channels 0-3 are enabled)
 - Or two (2) stereo I2S outputs (channels 2-5 – only when I2S channels 0-1 are enabled)
 - I2S data aligned in I2S format; Contact Zoran for left-justified format support
- **I2S audio lines (shared with DDX) can be used as inputs**
 - Six (6) channel I2S input (3 stereo I2S pairs), data aligned in I2S format; Contact Zoran for left-justified and right-justified formats support
- **One (1) S/PDIF output**
- **Audio decode performed in either/both the audio DSP and CPU**
 - Audio DSP allows for a significant level of audio post-processing
 - L/R downmix for standard stereo digital or line-level output
 - Algorithms available for the following:
 - Dolby® AC-3 Class A
 - MPEG audio Layer 1 (ISO-13818-3)
 - “Musicam” MPEG audio Layer 2 (ISO-13818-3)
 - MP3 MPEG audio Layer 3 (ISO-13838-3)
 - Tone generation
 - Post-processing 3D surround & Dialog Clarity (SRS TruSurroundHD™, QSurround)
 - Post-processing bass and treble control (Audyssey® ABX)
 - Post-processing automatic volume control (Audyssey® AVL)
 - Post-processing 5-band equalizer (Audyssey® AEQ)
 - Supports audio and video PTS synchronization
 - Stores processed streams in memory for playback using APU
- **Audio Processing Unit (APU)**
 - Single independent integrated APU unit
 - Audio playback from unified memory
 - Audio select, mix, cross-fade, and attenuate all audio sources
 - Supports multiple serial data formats
 - Supports sample rates up to 96 KHz
 - IEC-958 output of encoded or PCM audio data

1.2.6. Video Decoders

- **MPEG MP@HL decoder**
 - Decode of a single HD (MP@HL) stream
 - Decodes of ISO-13818-2 MP@ML, MP@HL
 - Decode of all ATSC-compliant formats
 - Slice-level and frame-level error concealment
 - The decoder engine can decode MPEG-compressed bitstreams as defined in the following specifications:
 - ISO/IEC 13818-2, "Information Technology - Generic Coding of Moving Pictures and Associated Audio Information: Video," (Up to MP@HL)
 - A/53, "ATSC Digital Television Standard," (Table 3)
 - DTVMDB04, "DIRECTV MPEG-2 Video Bitstream Specification for the IRD"
- **Integrated NTSC/PAL decoder**
 - 3D adaptive comb filter
 - Eliminates dot crawl from vertical or horizontal transitions
 - Eliminates dot crawl from single pixel lines
 - Eliminates false color from high frequency horizontal luma
 - Performs ideal YC separation for still image
 - No loss in horizontal or vertical chroma detail
 - No loss in horizontal or vertical luma detail
 - Performs well both on real video images and on test patterns
 - Adaptive horizontal PLL
 - Automatically adjusts loop bandwidth for signal conditions
 - Automatically detects VCR source and enters optimum tracking mode; most decoders require a "VCR mode" bit to be set to optimally handle VCR signals
 - Automatically detects VCR special effects mode and compensates
 - Comb filter automatically disabled when VCR source is detected
 - Robust sync and DC setup acquisition
 - DC setup and sync recovery is very robust even in the presence of noise, ghosting, and unlock condition
 - Automatic switch over to "fine" mode operation once rough lock is acquired
 - Chroma edge enhancement
 - Improves the horizontal transition of the chroma edge
 - VBI decoder
 - Performs VBI data capture and data slicing embedded in the video lines (composite, S-Video, analog RF input)
- **JPEG decoding**

1.2.7. Front-End Demod / Demux

- **Integrated 8VSB/QAM-B demodulator**
 - ATSC 8-VSB demodulation
 - Enhanced 8-VSB multi-path performance with wide equalizer coverage
 - Superior VSB indoor reception using enhanced equalization and synchronization algorithms, enabling Brazil and other 0 dB ghost reception
 - Adaptive control loops dependent upon channel conditions for fast channel acquisition and optimal tracking
 - Advanced doppler ghost rejection
 - QAM-B demodulation
 - ANSI/SCTE 07, ITU-T J.83 Annex B 64- /256-QAM, 5.06/5.36 Msymbol/sec rate, respectively
 - Support all DI modes up to I=128, J=8
 - 84-tap equalization range: 36 FFE and 48 DFE for superior cable micro-reflections rejection
 - Enhanced phase noise rejection
 - Excellent burst noise and combined distortion rejection
 - Exceptional AM noise rejection
 - Fast channel auto search based on auto 64- /256-QAM detection and wide carrier acquisition range
 - Advanced system functions
 - Accepts 44 MHz from the tuner, eliminating external base-band demodulation
 - IF AGC PWM output
 - All digital recovery loops
 - FEC statistics, receiver status, and channel data such as S/N ratio, equalizer taps, carrier offset, and more are available
 - Adaptive selection of receiver
 - Adaptive recovery loops based on channel conditions are used to achieve optimum reception for both high doppler echoes conditions and 0dB conditions
 - The synchronization and the equalization algorithms are based on both training signals and blind data
 - It enables better channel tracking – resulting in achieving all A74 requirements
 - Fast channel acquisition in 0dB conditions, < 0.5sec.
 - Improves immunity to noise for Brazil ensembles over previous Zoran devices
 - Improved phase noise rejection in 0dB conditions
- **NTSC demodulator**
 - Fully programmable digital video frequency and group delay equalization including internal digital Nyquist filter and excellent sound carrier digital rejection (>60dB)
 - Digital carrier recovery (AFT) with accurate report to host
 - Digital carrier recovery without quadrature distortions
 - Excellent (110%) over modulation at all white signal (100IRE)
 - Digital video IF AGC and optional delayed tuner AGC with programmable take over point
 - AM interference rejection
- **BTSC/A2 demodulator**
 - BTSC mono, stereo and SAP DBX decoding for US NTSC TV reception
 - A2 mono, stereo and bilingual decoding for Korea NTSC TV reception
- **TS demultiplexer**
 - Maximum transport bitrate: 80 Mbit/sec
 - ISO-13818-1 compliant
 - Supports PID filtering - total number of simultaneous PID filters: 32
 - ATSC-compliant transport demultiplexer
 - Maximum filtered (output) demux bit rate of 80 Mbits/sec
 - PCR locking using internal STC counter and VCXO control
- **Demodulator inputs**
 - One (1) differential IF pair for all tuner formats
 - One (1) SIF (sound IF) for audio-only formats

1.2.8. Memory Support

- **16-bit DDR2 interface (400MHz or 475 MHz)**
 - Up to 1.87 GByte/second peak memory throughput
 - 400MHz DDR2 sufficient for WXGA designs
 - 400MHz DDR2 sufficient for 1080p designs without TCON /overdrive
 - 533MHz DDR2 (clocked at 475MHz) sufficient for 1080p designs with TCON /overdrive
 - Up to 128 MBytes maximum memory
 - Typical 64MByte system implementation for WXGA and 1080p designs
 - High performance arbiter with assignable client priorities
 - SSTL-18 Class 1 electrical interface
- **Serial FLASH**
 - 40MHz SPI clock
 - Up to 16 MBytes maximum memory
 - Typical 2-4 MByte system implementation

1.2.9. Integrated TV MicroController

- **Support for “Sleep” mode operation**
- **Front panel I/O support (buttons and display)**
- **IRR input**
- **General-purpose 8-bit ADC with 5 multiplexed inputs**
 - i.e. Voltage monitoring
- **Sleep timer**
- **Watchdog timer**
- **GPIO interrupt control**
- **Support for A/V input monitoring**
 - Monitors the HDMI inputs for activity
- **Integrated EDID memory for HDMI inputs and VGA inputs**
 - 512 bytes memory x 4 input ports
- **Support for automatic VGA signal detection and wake-up**
- **HDMI CEC support**
- **UART for debug**
- **Real-time clock support**

1.2.10. Graphics Processing

- **32-bit RGB / YCbCr**
- **16-bit RGB**
- **8-bit indexed with CLUT**
- **Graphics Block Transfer (BLT)**
 - Supports copy, bit depth conversion and alpha blending of 8-, 16- and 32-bit pixel maps with 32-bit output
 - Supports Porter-Duff alpha blending formulas
 - Alpha destination and alpha compare
 - Point, Line, Rectangle, Text and Trapezoid Draw functions
 - Rectangle Fill function



- **Graphics Unit Scaler (GUS)**
 - Support scaling and blending of several graphics planes in a single operation
 - Can also perform simple BLT operations (BitBlt, stretch BitBlt, trapezoid BitBlt, mirror BitBlt, rotate BitBlt)
- **Color space converter**
- **Raster Operation (ROP)**

1.2.11. System Interfaces

- **Two (2) PWM outputs**
- **Three (3) 2-signal UARTs**
 - Maximum baud rate: 115200
 - 16550 compatible
 - Third UART is allocated to TVuC and shared with main CPU UART
- **Two (2) I2C master or slave interfaces**
 - Maximum bitrate: 400 Kb/s
 - Master or slave mode
- **One (1) IR receiver, with hardware demodulation**
- **SPI interface**
 - Up to 40 MHz clock rate
 - Supports serial FLASH up to 16 MByte
 - Two (2) select signals for peripheral support
- **Integrated USB interface**
 - One High Speed USB v2.0 port

1.2.12. Security Features

- **Integrated One Time Programmable memory (OTP)**
 - 8 Kb of One Time Programmable (OTP) secure memory
 - Used for secure storage:
 - HDCP Key Selection Vectors (KSVs)
 - Error Correction (ECC) Checksum and data
 - Readable ONLY by specific IROM instructions programmed into the SupraHD® 748
 - HDMI keys are encrypted with a proprietary Zoran encryption algorithm during the programming process

1.2.13. Misc. IC Information

- **25.000 MHz crystal input required to support standard ATSC timing**

1.3. SupraHD® 748 Block Diagrams

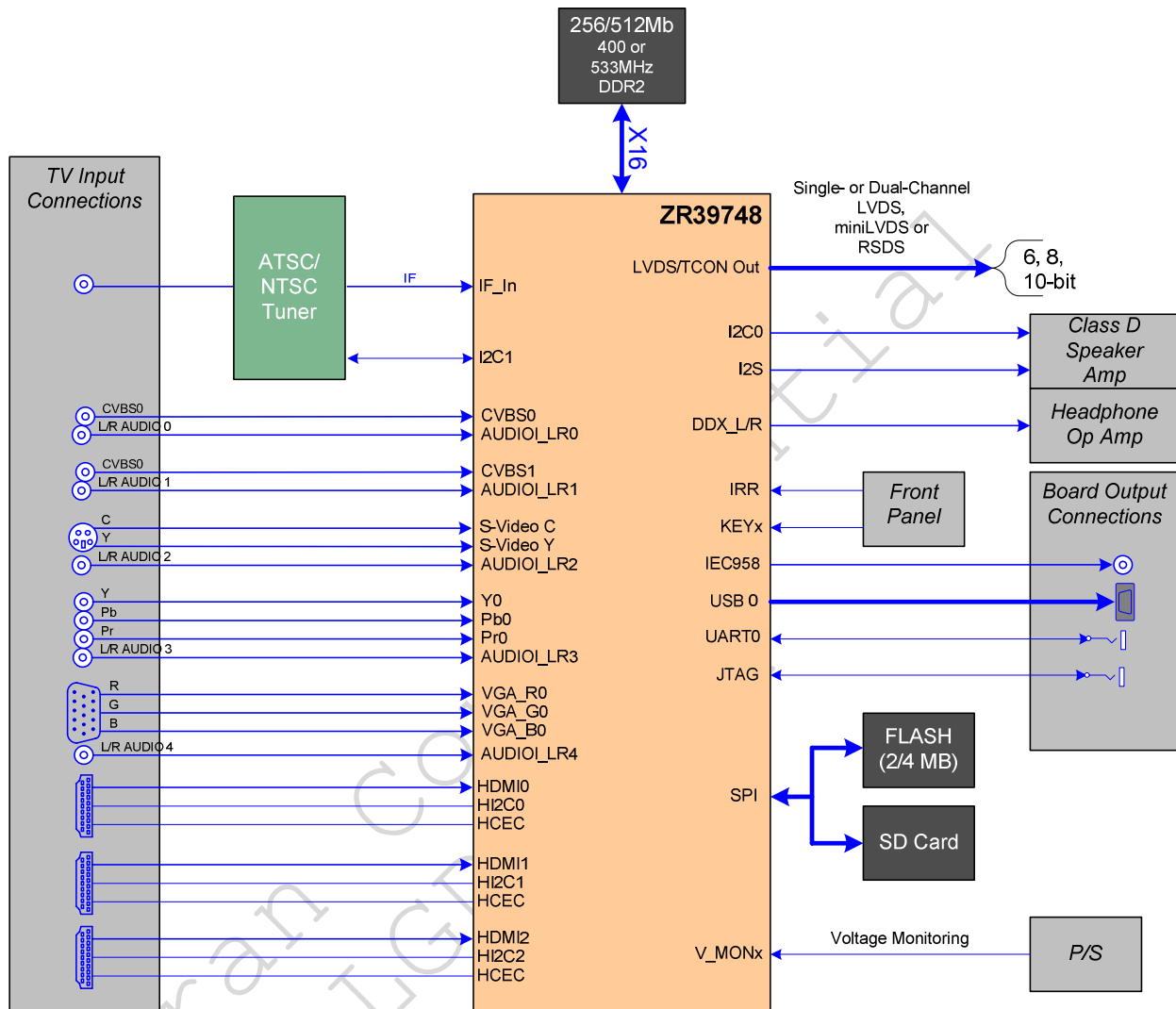


Figure 2: SupraHD® 748 Typical ATSC TV System Implementation Block Diagram

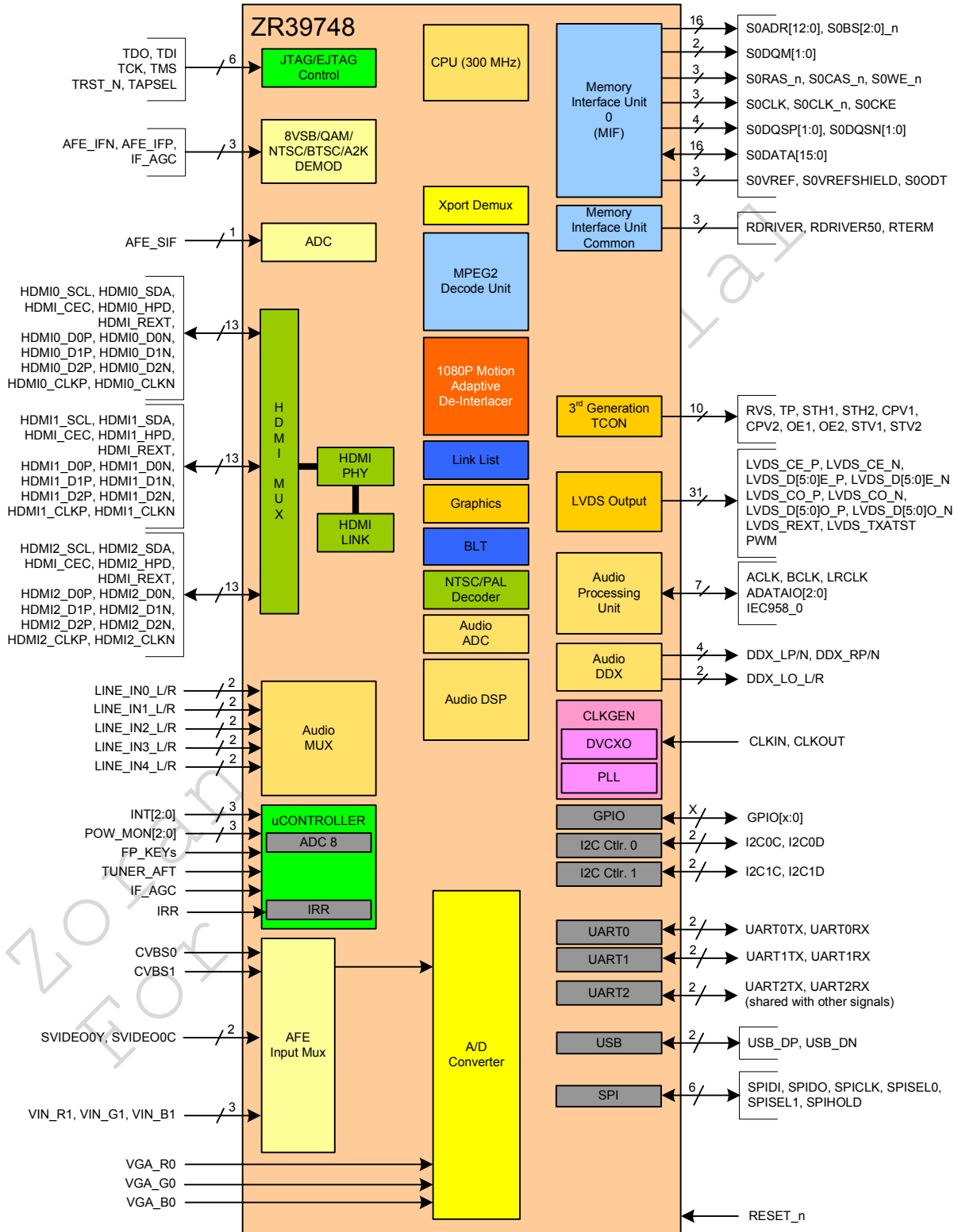
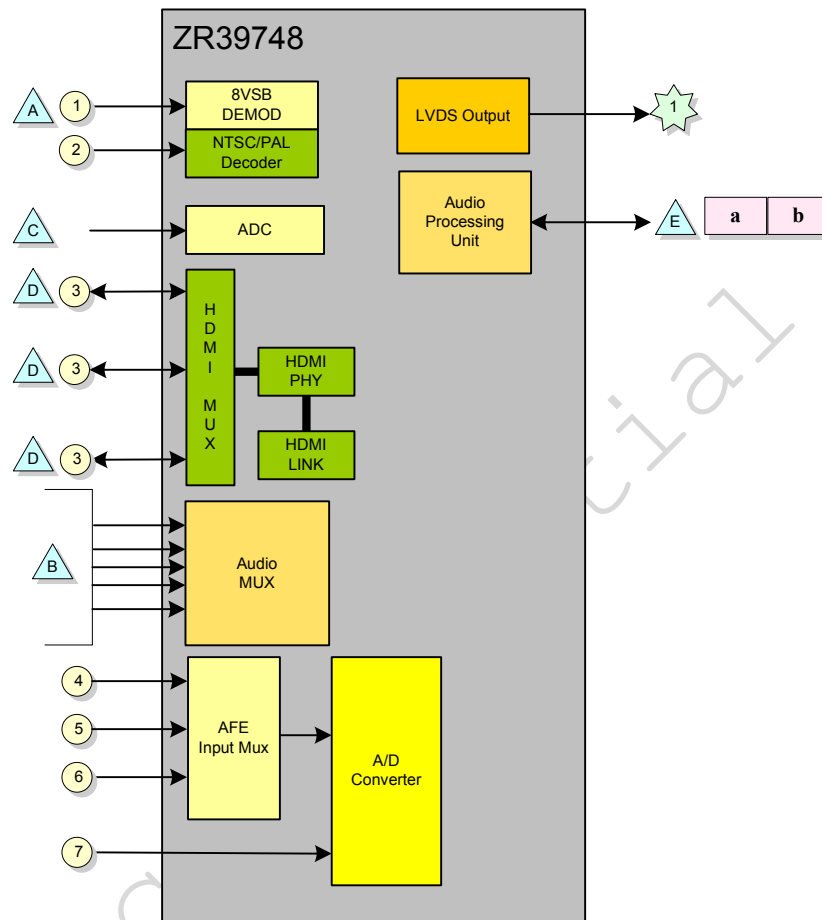


Figure 3: SupraHD® 748 Detailed Block Diagram



Video Input Options

1. MPEG2 Transport Stream Input via IF from Tuner
2. NTSC Video from Tuner
3. Uncompressed Digital Video (YUV or RGB Format) via HDMI Port (x3)
4. CVBS SD Analog Video (x2)
5. S-Video SD Analog Video
6. HD Analog Video from YPbPr Port
7. HD Analog Video from RGB Port

Video Output Options

1. Uncompressed Digital Video (YUV or RGB Format) via LVDS Output

Audio Input Options

- A. MPEG2 Transport Stream Input via IF from Tuner
- B. L/R Line-Level Analog Inputs
- C. SIF Audio from NTSC Tuner
- D. Dolby Digital or two (2) channel PCM Audio Input via the HDMI Port
- E. Up to Six (6) channel I2S Input via three (3) I2S Ports

Audio Output Options

- a. Up to Six (6) channel I2S Output via three (3) I2S Ports, IEC958 Output
- b. Stereo L/R DDX Output (speakers) and Stereo L/R Line Out

Figure 4: SupraHD® 748 I/O Option Dataflows

Chapter 2. Signal Listings

2.1. BGA Package Signal Lists

The following tables list the signals in alpha-numeric ball order, by signals that have one or more alternate functions and by functional group.

2.1.1. Signal Listing by Ball Order

The following table lists the BGA package signals in ball order. The following information is provided for each signal:

- Ball number
- Signal name - primary function for signals with multiple function options
- I/O type
 - I = digital input, O = digital output, B = digital input/output
 - AI = analog input, AO = analog output, AI/O = analog input/output, P = power/ground
- Indication if the signal includes an internal pull-up or pull-down resistor
 - P.U. = pull-up, P.D. = pull-down, x = not applicable
- Default state of the signal immediately after RESET (after /RESET_n is pulled high)
 - I = input, O = output

Table 1: SupraHD® 748 BGA Signals – Ball Order

Ball #	Signal Name	I/O Type	Internal pull-up / down?	Default after RESET
A1	HDMI_CEC	B (open drain)	x	I
A2	IRR	I	x	I
A3	VGA_SDA	B (open drain)	x	I
A4	IF_AGC	O (open drain)	x	Low
A5	BOOT_OPT/LED	B	x	I
A6	UART0TX	B (open drain)	x	O (High)
A7	CLKIN_RTC	AI	x	x
A8	CLKOUT_25M	AO	x	x
A9	DDX_LO_L	B	PU	O (Low)
A10	DDX_LP	B	PU	O (Low)
A11	DDX_RP	B	PU	O (Low)
A12	AFE_IFP	AI	x	x
A13	AFE_SIF	AI	x	x
A14	AFE_CVBS2	AI	x	x
A15	AFE_SVID_Y	AI	x	x
A16	AFE_SVID_C	AI	x	x
A17	AC_L4	AI	x	x
A18	AC_R5	AI	x	x

Table 1: SupraHD® 748 BGA Signals – Ball Order

Ball #	Signal Name	I/O Type	Internal pull-up / down?	Default after RESET
A19	AC_R3	AI	x	x
A20	AC_R1	AI	x	x
A21	I2C1C	B (open drain)	x	I
A22	I2C1D	B	x	I
B1	GPIO_TV_P7	B	x	I
B2	INT2	B	x	I
B3	VGA_SCL	B (open drain)	x	I
B4	T0	B	PU	I
B5	WDT_EN_N	B	x	I
B6	T1	B	PU	I
B7	CLKOUT_RTC	AO	x	x
B8	CLKIN_25M	AI	x	x
B9	DDX_LN	B	PU	O (Low)
B10	DDX_RN	B	PU	O (Low)
B11	AFE_SOY	AI	x	x
B12	AFE_IFN	AI	x	x
B13	AFE_CVBS1	AI	x	x
B14	AFE_VGA_R	AI	x	x
B15	AFE_VGA_G	AI	x	x
B16	AFE_VGA_B	AI	x	x
B17	AC_L3	AI	x	x
B18	AC_L5	AI	x	x
B19	AC_R4	AI	x	x
B20	AC_R2	AI	x	x
B21	SPI_CLK	O	x	Low
B22	SPI_DI	I	x	I
C1	GPIO_TV_P3	B	x	I
C2	GPIO_TV_P6	B	x	I
C3	GPIO_TV_P5	B	x	I
C4	GPIO_TV_P4	B	x	I
C5	POWER_CTL2	B	x	O (High)
C6	INT0_N	B	x	I
C7	UART0RX	B (open drain)	x	I
C8	ADC8_IN0	AI	x	x
C9	ADC8_IN2	AI	x	x
C10	DDX_LO_R	B	PU	O (Low)
C11	AFE_SOG	AI	x	x
C12	AFE_FILTOUT	AO	x	x
C13	AFE_PR	AI	x	x
C14	AFE_Y	AI	x	x
C15	AFE_PB	AI	x	x
C16	AC_L1	AI	x	x
C17	AC_L2	AI	x	x

Table 1: SupraHD® 748 BGA Signals – Ball Order

Ball #	Signal Name	I/O Type	Internal pull-up/down?	Default after RESET
C18	TMS	B	PU	I
C19	IEC958O	O	x	O (Low)
C20	SPI_HOLD	B	PU	O (High)
C21	SPI_SEL0	O	x	High
C22	SPI_DO	B	x	O (Low)
D1	HDMI0_5VSNS	B	x	I
D2	GPIO_TV_P2	B	x	I
D3	GPIO_TV_P1	B	x	I
D4	GPIO_TV_P0	B	x	I
D5	POWER_CTL1	B	x	O (High)
D6	INT1_N	B	x	I
D7	TVCPU_PWM0	O	x	Low
D8	ADC8_IN1	AI	x	x
D9	ADC8_IN3	AI	x	x
D10	AFE_FILTOUT2	AO	x	x
D11	AFE_VPWR33_PLL	P	x	x
D12	AFE_RSET	AI	x	x
D13	ADC8_IN6	AI	x	x
D16	AC_RCAP_N	AO	x	x
D17	AC_RCAP_P	AO	x	x
D18	TDO	B	x	I
D19	TDI	B	PU	I
D20	TCK	B	PU	I
D21	LVDS_CUP	AO	x	x
D22	LVDS_CUN	AO	x	x
E1	RESET_N	I	x	I
E2	HDMI0_HPD	B	x	I
E3	HDMI0_SDA	B (open drain)	x	I
E4	HDMI0_SCL	B (open drain)	x	I
E20	UART1RX	I	PU	I
E21	LVDS_D0UP	AO	x	x
E22	LVDS_D0UN	AO	x	x
F1	GPIO_P4	B	PU	I
F2	GPIO_P0	B	PD	I
F3	I2C0D	B (open drain)	x	I
F4	I2C0C	B (open drain)	x	I
F6	ADC8_RBIAS	AI	x	x
F7	CLK25M_VDD	P	x	x
F8	DDX_VDDIO	P	x	x
F9	AUDPLL_VDDIO	P	x	x
F10	AFE_VPWR10_PLL	P	x	x
F11	AFE_VPWR33_REF	P	x	x
F12	AFE_VPWR33	P	x	x

Table 1: SupraHD® 748 BGA Signals – Ball Order

Ball #	Signal Name	I/O Type	Internal pull-up/down?	Default after RESET
F13	AFE_VPWR10	P	x	x
F14	AC_LCAP_P	AO	x	x
F15	AC_LCAP_N	AO	x	x
F16	AC_AVDD	P	x	x
F17	AC_RES	AO	x	x
F18	AC_VCM	AO	x	x
F20	UART1TX	O	PU	High
F21	LVDS_D1UP	AO	x	x
F22	LVDS_D1UN	AO	x	x
G1	GPIO_P8	B	PU	I
G2	GPIO_P7	B	PU	I
G3	GPIO_P6	B	PU	I
G4	GPIO_P5	B	PU	I
G6	IOVDD_STBY	P	x	x
G7	COREVDD_STBY	P	x	x
G8	VDDIO_SYSPLL	P	x	x
G17	AC_VREFP	AO	x	x
G18	AC_VREFN	AO	x	x
G20	GND	P	x	x
G21	LVDS_D2UP	AO	x	x
G22	LVDS_D2UN	AO	x	x
H1	GPIO_P12	B	PU	I
H2	GPIO_P11	B	x	I
H3	GPIO_P10	B	PU	I
H4	GPIO_P9	B	PU	I
H6	IOVDD_STBY	P	x	x
H9	AUDPLL_GND	P	x	x
H10	GNDSYSPLL	P	x	x
H11	AFE_PLL_GND	P	x	x
H12	AFE_GND	P	x	x
H13	GND	P	x	x
H14	GND	P	x	x
H15	AC_AGNDREF	P	x	x
H17	CORE_VDD	P	x	x
H20	LVDS_D5UP	AO	x	x
H21	LVDS_D3UP	AO	x	x
H22	LVDS_D3UN	AO	x	x
J1	HDMI2_CN	AI	x	x
J2	HDMI2_CP	AI	x	x
J3	AFE_HSYNC_IN	B	x	I
J4	GPIO_P14	B	PU	I
J6	CORE_VDD	P	x	x
J8	GND	P	x	x

Table 1: SupraHD® 748 BGA Signals – Ball Order

Ball #	Signal Name	I/O Type	Internal pull-up/down?	Default after RESET
J9	GND	P	x	x
J10	GND	P	x	x
J11	GND	P	x	x
J12	GND	P	x	x
J13	GND	P	x	x
J14	GND	P	x	x
J15	GND	P	x	x
J20	LVDS_D5UN	AO	x	x
J21	LVDS_D4UP	AO	x	x
J22	LVDS_D4UN	AO	x	x
K1	HDMI2_D0N	AI	x	x
K2	HDMI2_D0P	AI	x	x
K3	GND	P	x	x
K8	GND	P	x	x
K9	GND	P	x	x
K10	GND	P	x	x
K11	GND	P	x	x
K12	GND	P	x	x
K13	GND	P	x	x
K14	GND	P	x	x
K15	GND	P	x	x
K17	IOVDD	P	x	x
K18	IOVDD	P	x	x
K20	GND	P	x	x
K21	LVDS_CLP	AO	x	x
K22	LVDS_CLN	AO	x	x
L1	HDMI2_D1N	AI	x	x
L2	HDMI2_D1P	AI	x	x
L3	GPIO_P1	B	PU	I
L5	IOVDD	P	x	x
L6	IOVDD	P	x	x
L8	GND	P	x	x
L9	GND	P	x	x
L10	GND	P	x	x
L11	GND	P	x	x
L12	GND	P	x	x
L13	GND	P	x	x
L14	GND	P	x	x
L15	LVDS_ATEST	AO	x	x
L17	LDI_VDD	P	x	x
L18	LDI_VDD	P	x	x
L20	GND	P	x	x
L21	LVDS_D0LP	AO	x	x

Table 1: SupraHD® 748 BGA Signals – Ball Order

Ball #	Signal Name	I/O Type	Internal pull-up/down?	Default after RESET
L22	LVDS_D0LN	AO	x	x
M1	HDMI2_D2N	AI	x	x
M2	HDMI2_D2P	AI	x	x
M3	GND	P	x	x
M8	GND	P	x	x
M9	GND	P	x	x
M10	GND	P	x	x
M11	GND	P	x	x
M12	GND	P	x	x
M13	GND	P	x	x
M14	GND	P	x	x
M15	GND	P	x	x
M20	GND	P	x	x
M21	LVDS_D1LP	AO	x	x
M22	LVDS_D1LN	AO	x	x
N1	HDMI1_CN	AI	x	x
N2	HDMI1_CP	AI	x	x
N3	GPIO_P16	B	PU	I
N5	HDMI_VDD3P3	P	x	x
N6	HDMI_VDD3P3	P	x	x
N8	GND	P	x	x
N9	GND	P	x	x
N10	GND	P	x	x
N11	GND	P	x	x
N12	GND	P	x	x
N13	GND	P	x	x
N14	GND	P	x	x
N15	GND	P	x	x
N17	LDI_CVDD	P	x	x
N18	LVDS_REXT	AI	x	x
N20	LVDS_D5LP	AO	x	x
N21	LVDS_D2LP	AO	x	x
N22	LVDS_D2LN	AO	x	x
P1	HDMI1_D0N	AI	x	x
P2	HDMI1_D0P	AI	x	x
P3	GND	P	x	x
P8	GND	P	x	x
P9	GND	P	x	x
P10	GND	P	x	x
P11	GND	P	x	x
P12	GND	P	x	x
P13	GND	P	x	x
P14	GND	P	x	x

Table 1: SupraHD® 748 BGA Signals – Ball Order

Ball #	Signal Name	I/O Type	Internal pull-up/down?	Default after RESET
P15	GND	P	x	x
P20	LVDS_D5LN	AO	x	x
P21	LVDS_D3LP	AO	x	x
P22	LVDS_D3LN	AO	x	x
R1	HDMI1_D1N	AI	x	x
R2	HDMI1_D1P	AI	x	x
R3	GND	P	x	x
R5	CORE_VDD	P	x	x
R6	HDMI_VDD1V	P	x	x
R8	HDMI_VDD3P3	P	x	x
R9	GND	P	x	x
R10	CVSS_MSD	P	x	x
R11	CVSS_MSD	P	x	x
R12	CVSS_MSD	P	x	x
R13	CVSS_MSD	P	x	x
R14	GND	P	x	x
R15	DDRPLL_GND	P	x	x
R17	CORE_VDD	P	x	x
R18	CORE_VDD	P	x	x
R20	GND	P	x	x
R21	LVDS_D4LP	AO	x	x
R22	LVDS_D4LN	AO	x	x
T1	HDMI1_D2N	AI	x	x
T2	HDMI1_D2P	AI	x	x
T3	GND	P	x	x
T5	CORE_VDD	P	x	x
T6	CORE_VDD	P	x	x
T17	USB2_VDDA33T	P	x	x
T18	USB_REXT	AI	x	x
T20	TCON_GPO	B	PU	I
T21	TCON_STV1	B	PU	I
T22	TCON_OE1	B	PU	I
U1	HDMI0_CN	AI	x	x
U2	HDMI0_CP	AI	x	x
U3	HDMI_ATEST	AO	x	x
U5	HDMI_REXT	AI	x	x
U9	MVDD	P	x	x
U11	CVDD_MSD	P	x	x
U13	MVDD	P	x	x
U14	MVDD	P	x	x
U16	DDRPLL_AVDD	P	x	x
U17	RDRIVER50	AI	x	x
U20	TCON_STV2	B	PU	I



SupraHD®-748-DTS-GEN-v1.1

Table 1: SupraHD® 748 BGA Signals – Ball Order

Ball #	Signal Name	I/O Type	Internal pull-up/down?	Default after RESET
U21	TCON_STH1	B	PU	I
U22	TCON_STH2	B	PU	I
V1	HDMI0_D0N	AI	x	x
V2	HDMI0_D0P	AI	x	x
V3	GND	P	x	x
V8	MVDD	P	x	x
V9	MVDD	P	x	x
V11	CVDD_MSD	P	x	x
V13	S0VREF	AI	x	x
V14	MVDD	P	x	x
V16	RTERM	AI	x	x
V17	RDRIVER	AI	x	x
V19	USB_ATEST	AO	x	x
V20	TCON_CPV1	B	PU	I
V21	TCON_OE2	B	PU	I
V22	TCON_TP	B	PU	I
W1	HDMI0_D1N	AI	x	x
W2	HDMI0_D1P	AI	x	x
W3	GND	P	x	x
W19	TCON_RVS	B	PU	I
W20	TCON_CPV2	B	PU	I
W21	USB_DN	AI/O	x	x
W22	USB_DP	AI/O	x	x
Y1	HDMI0_D2N	AI	x	x
Y2	HDMI0_D2P	AI	x	x
Y3	GND	P	x	x
Y4	GND	P	x	x
Y5	GND	P	x	x
Y6	GND	P	x	x
Y7	GND	P	x	x
Y8	GND	P	x	x
Y9	GND	P	x	x
Y10	GND	P	x	x
Y11	GND	P	x	x
Y12	GND	P	x	x
Y13	GND	P	x	x
Y14	GND	P	x	x
Y15	GND	P	x	x
Y16	GND	P	x	x
Y17	GND	P	x	x
Y18	GND	P	x	x
Y19	GND	P	x	x
Y20	PWM	O	x	Low



SupraHD®-748-DTS-GEN-v1.1

Table 1: SupraHD® 748 BGA Signals – Ball Order

Ball #	Signal Name	I/O Type	Internal pull-up / down?	Default after RESET
Y21	SPI_SEL1	B	PU	O (High)
Y22	S0ADR11	O	x	Low
AA1	S0ADR9	O	x	Low
AA2	S0ADR5	O	x	Low
AA3	S0ADR1	O	x	Low
AA4	S0BS1	O	x	Low
AA5	S0BS2	O	x	Low
AA6	S0CKE	O	x	Low
AA7	S0DAT4	B	x	I
AA8	S0DAT6	B	x	I
AA9	S0DAT14	B	x	I
AA10	S0DAT9	B	x	I
AA11	S0DQM0	O	x	Low
AA12	S0DAT2	B	x	I
AA13	S0DAT7	B	x	I
AA14	S0DQS1P	B	x	I
AA15	S0DAT15	B	x	I
AA16	S0DAT13	B	x	I
AA17	S0DQS0P	B	x	I
AA18	S0CLKN	B	x	I
AA19	S0RASN	O	x	Low
AA20	S0ADR0	O	x	Low
AA21	S0ADR4	O	x	Low
AA22	S0ADR8	O	x	Low
AB1	S0ADR12	O	x	Low
AB2	S0ADR7	O	x	Low
AB3	S0ADR3	O	x	Low
AB4	S0ADR10	O	x	Low
AB5	S0BS0	O	x	Low
AB6	S0WEN	O	x	Low
AB7	S0DAT3	B	x	I
AB8	S0DAT1	B	x	I
AB9	S0DAT12	B	x	I
AB10	S0DQM1	O	x	Low
AB11	S0DAT11	B	x	I
AB12	S0DAT5	B	x	I
AB13	S0DAT0	B	x	I
AB14	S0DAT10	B	x	I
AB15	S0DQS1N	B	x	I
AB16	S0DAT8	B	x	I
AB17	S0DQS0N	B	x	I
AB18	S0CLKP	O	x	High
AB19	S0ODT	O	x	Low



SupraHD®-748-DTS-GEN-v1.1

Table 1: SupraHD® 748 BGA Signals – Ball Order

Ball #	Signal Name	I/O Type	Internal pull-up/down?	Default after RESET
AB20	S0CASN	O	x	Low
AB21	S0ADR2	O	x	Low
AB22	S0ADR6	O	x	Low

Zoran Confidential
For LGD Only

2.1.2. Signals with Alternate Functions

The following table lists the BGA package signals that offer alternate functionality. The signals are listed in ball order. Signals not listed do not have any alternate functionality. The following information is provided for each signal:

- Ball number
- Signal name / 1st function – this is the primary function of the signal
- 2nd function – this is the secondary function of the signal
- 3rd function – this is the tertiary function of the signal
- 4th function – this is the fourth function of the signal

Function columns with blank entries in the 3rd and/or 4th function columns indicate that additional functionality is not applicable.

Table 2: SupraHD® 748 BGA Signals with Alternate Functions

Ball #	Signal Name / 1 st Function	2 nd Function	3 rd Function	4 th Function
A3	VGA_SDA	Reserved for debug		
A5	BOOT_OPT/LED	GPIO_TV_P26	Reserved for debug	
A6	UART0TX	TVU_UART_TXD		
A9	DDX_LO_L	ACLK	DDX_5a	Reserved for debug
A10	DDX_LP	BCLK	DDX_3a	Reserved for debug
A11	DDX_RP	LRCLK	DDX_1a	Reserved for debug
A13	AFE_SIF	AFE_PR2		
A15	AFE_SVID_Y	AFE_Y2		
A16	AFE_SVID_C	AFE_PB2		
A21	I2C1C	GPIO_S[4]		
A22	I2C1D	GPIO_S[5]		
B1	GPIO_TV_P7	HDMI2_HPDP		
B2	INT2	AFE_VSYNC_IN	GPIO_TV_P29	Reserved for debug
B3	VGA_SCL	Reserved for debug		
B4	T0	GPIO_TV_P21	Reserved for debug	
B5	WDT_EN_N	GPIO_TV_P27	T2	Reserved for debug
B6	T1	GPIO_TV_P24	Reserved for debug	
B9	DDX_LN	ADATAIO[0]	DDX_4a	Reserved for debug
B10	DDX_RN	ADATAIO[2]	DDX_2a	Reserved for debug
B21	SPI_CLK	GPIO_S[14]		
B22	SPI_DI	GPIO_S[7]		
C1	GPIO_TV_P3	HDMI1_5VSENSE		
C2	GPIO_TV_P6	HDMI2_5VSENSE		
C3	GPIO_TV_P5	HDMI2_SDA		
C4	GPIO_TV_P4	HDMI2_SCL		
C5	POWER_CTL2	GPIO_TV_P12	Reserved for debug	
C6	INT0_N	GPIO_TV_P13	Reserved for debug	
C7	UART0RX	TVU_UART_RXD		
C10	DDX_LO_R	ADATAIO[1]	DDX_6a	Reserved for debug
C18	TMS	EJTS	Reserved for debug	



Table 2: SupraHD® 748 BGA Signals with Alternate Functions

Ball #	Signal Name / 1 st Function	2 nd Function	3 rd Function	4 th Function
C19	IEC958O	GPIO_S[10]	Reserved for debug	
C20	SPI_HOLD	GPIO_S[12]		
C21	SPI_SEL0	GPIO_S[15]		
C22	SPI_DO	GPIO_S[13]		
D2	GPIO_TV_P2	HDMI1_HPD		
D3	GPIO_TV_P1	HDMI1_SDA		
D4	GPIO_TV_P0	HDMI1_SCL		
D5	POWER_CTL1	GPIO_TV_P15	Reserved for debug	
D6	INT1_N	GPIO_TV_P23	Reserved for debug	
D18	TDO	EJTDO	Reserved for debug	
D19	TDI	EJTDI	Reserved for debug	
D20	TCK	EJTCK	Reserved for debug	
E20	UART1RX	GPIO_S[3]		
F1	GPIO_P4	ACLK	Reserved for debug	
F2	GPIO_P0	BOOT_0	LRCLK	Reserved for debug
F20	UART1TX	GPIO_S[2]		
G1	GPIO_P8	Reserved for debug		
G2	GPIO_P7	Reserved for debug		
G3	GPIO_P6	Reserved for debug		
G4	GPIO_P5	ADATAIO[0]	Reserved for debug	
H1	GPIO_P12	Reserved for debug		
H2	GPIO_P11	PWM2	Reserved for debug	
H3	GPIO_P10	Reserved for debug		
H4	GPIO_P9	Reserved for debug		
J3	AFE_HSYNC_IN	Reserved for debug		
J4	GPIO_P14	Reserved for debug		
L3	GPIO_P1	BOOT_1	BCLK	Reserved for debug
N3	GPIO_P16	Reserved for debug		
T20	TCON_GPO	Reserved for debug		
T21	TCON_STV1	GPIO_S[9]		
T22	TCON_OE1	GPIO_S[8]		
U20	TCON_STV2	Reserved for debug		
U21	TCON_STH1	Reserved for debug		
U22	TCON_STH2	Reserved for debug		
V20	TCON_CPV1	Reserved for debug		
V21	TCON_OE2	Reserved for debug		
V22	TCON_TP	TAPSEL	Reserved for debug	
W19	TCON_RVS	Reserved for debug	Reserved for debug	
W20	TCON_CPV2	TAPSEL_CAS	Reserved for debug	
Y20	PWM	GPIO_S[16]		
Y21	SPI_SEL1	GPIO_S[11]		

2.1.3. Signal Listings by Function

The following tables list the BGA package signals grouped according to function. The following information is provided for each signal:

- Ball number – numbers with a preceding (*) indicate 2nd / 3rd / 4th functionality for the signal
- Signal/function name - can be the primary, secondary, tertiary or 4th function of the signal
- I/O type
 - I = digital input, O = digital output, B = digital input/output
 - AI = analog input, AO = analog output, AI/O = analog input/output, P = power/ground
- Function description

Table 3: SupraHD® 748 BGA Signals – JTAG/EJTAG

Ball #	Signal / Function	I/O Type	Description
(*) V22	TAPSEL	B	EJTAG Select This signal determines the functionality of the JTAG/EJTAG signals: When low, JTAG functionality is enabled. When high, EJTAG functionality is enabled. Default is EJTAG.
D20	TCK	B	JTAG Clock
D19	TDI	B	JTAG Data In
D18	TDO	B	JTAG Data Out
C18	TMS	B	JTAG Mode Select
(*) W20	TAPSEL_CAS	B	Test Signal
(*) D20	EJTCK	B	EJTAG Clock
(*) D19	EJTDI	B	EJTAG Data In
(*) D18	EJTDO	B	EJTAG Data Out
(*) C18	EJTMS	B	EJTAG Mode Select

Table 4: SupraHD® 748 BGA Signals – Analog Video Front-End Input

Ball #	Signal / Function	I/O Type	Description
A12	AFE_IFP	AI	Analog IF positive Differential input. Analog input to Main ADC. This input has built-in ESD protection rated up to 2000V. Input resistance - 25 KOhm
B12	AFE_IFN	AI	Analog IF Negative Differential input. Analog input to Main ADC. This input has built-in ESD protection rated up to 2000V. Input resistance - 25 KOhm
A13	AFE_SIF	AI	SIF (Sound IF) Input Analog input to to SIF ADC. This input has built-in ESD protection rated up to 2000V. Input resistance - 25 KOhm
A4	IF_AGC	O (open drain)	IF AGC Signal
B13	AFE_CVBS1	AI	Composite Video Input 1 This input has built-in ESD protection rated up to 2000V.
A14	AFE_CVBS2	AI	Composite Video Input 2 This input has built-in ESD protection rated up to 2000V.

Table 4: SupraHD® 748 BGA Signals – Analog Video Front-End Input

Ball #	Signal / Function	I/O Type	Description
A15	AFE_SVID_Y	AI	S-Video Y (Luma) Input This input has built-in ESD protection rated up to 2000V.
A16	AFE_SVID_C	AI	S-Video C (Chroma) Input This input has built-in ESD protection rated up to 2000V.
C14	AFE_Y	AI	Component Y Input This input has built-in ESD protection rated up to 2000V.
C15	AFE_PB	AI	Component Pb Input This input has built-in ESD protection rated up to 2000V.
C13	AFE_PR	AI	Component Pr Input This input has built-in ESD protection rated up to 2000V.
(*) A15	AFE_Y2	AI	2nd Component Y Input This input has built-in ESD protection rated up to 2000V.
(*) A16	AFE_PB2	AI	2nd Component Pb Input This input has built-in ESD protection rated up to 2000V.
(*) A13	AFE_PR2	AI	2nd Component Pr Input This input has built-in ESD protection rated up to 2000V.
B11	AFE_SOY	AI	Sync-On-Y (Luma) Input
B14	AFE_VGA_R	AI	VGA Red Input This input has built-in ESD protection rated up to 2000V.
B15	AFE_VGA_G	AI	VGA Green Input This input has built-in ESD protection rated up to 2000V.
B16	AFE_VGA_B	AI	VGA Blue Input This input has built-in ESD protection rated up to 2000V.
C11	AFE_SOG	AI	Sync-On-Green Input
J3	AFE_HSYNC_IN	B	Horizontal Sync Input for VGA
(*) B2	AFE_VSYNC_IN	B	Vertical Sync Input for VGA
B3	VGA_SCL	B (open drain)	VGA I2C Clock I2C clock for DDC information. This signal should be pulled high with a 47K ohm resistor (to 5V_VGA on the VGA connector or to 5V standby power).
A3	VGA_SDA	B (open drain)	VGA I2C Data I2C data for DDC information This signal should be pulled high with a 47K ohm resistor (to 5V_VGA on the VGA connector or to 5V standby power).
C12	AFE_FILTOUT	AO	AFE PLL Filter Filter connection to R/C filter. Recommended: 200pF cap to AFE 1.1V.
D10	AFE_FILTOUT2	AO	AFE PLL Filter 2 Filter connection to R/C filter. This signal includes an internal, programmable series resistor. Recommended: 3.9nF cap to AFE 1.1V
D12	AFE_RSET	AI	ADC Current Reference Setting This signal provides a current reference to the video ADCs. This signal has built-in ESD protection rated up to 2000V. Recommended: 0.1uF cap to GND.

Table 5: SupraHD® 748 BGA Signals – HDMI Input

Ball #	Signal / Function	I/O Type	Description
A1	HDMI_CEC	B (open drain)	HDMI Consumer Electronics Control
E2	HDMI0_HPD	B	HDMI0 Hot Plug Detect This signal is driven by the display device to indicate the presence of the display.
D1	HDMI0_5VSNS	B	HDMI0 5 Volt Sense Input
E4	HDMI0_SCL	B (open drain)	HDMI0 Serial Clock This signal is used to assert the data on HDMI_SDA. This signal follows the I2C protocol. This signal should be pulled high with a 47K ohm resistor (to 5V_SENSE on the HDMI connector or to 5V standby power).
E3	HDMI0_SDA	B (open drain)	HDMI0 Serial Data The HDMI interface sends and receives data on this I2C-compliant signal. This signal should be pulled high with a 47K ohm resistor (to 5V_SENSE on the HDMI connector or to 5V standby power).
U2	HDMI0_CP	AI	HDMI0 Clock - Positive
U1	HDMI0_CN	AI	HDMI0 Clock - Negative
Y2	HDMI0_D2P	AI	HDMI0 Data2 - Positive
Y1	HDMI0_D2N	AI	HDMI0 Data2 - Negative
W2	HDMI0_D1P	AI	HDMI0 Data1 - Positive
W1	HDMI0_D1N	AI	HDMI0 Data1 - Negative
V2	HDMI0_D0P	AI	HDMI0 Data0 - Positive
V1	HDMI0_D0N	AI	HDMI0 Data0 - Negative
(*) D2	HDMI1_HPD	B	HDMI1 Hot Plug Detect This signal is driven by the display device to indicate the presence of the display.
(*) C1	HDMI1_5VSENSE	B	HDMI1 5 Volt Sense Input
(*) D4	HDMI1_SCL	B	HDMI1 Serial Clock This signal is used to assert the data on HDMI_SDA. This signal follows the I2C protocol. This signal should be pulled high with a 47K ohm resistor (to 5V_SENSE on the HDMI connector or to 5V standby power).
(*) D3	HDMI1_SDA	B	HDMI1 Serial Data The HDMI interface sends and receives data on this I2C-compliant signal. This signal should be pulled high with a 47K ohm resistor (to 5V_SENSE on the HDMI connector or to 5V standby power).
N2	HDMI1_CP	AI	HDMI1 Clock - Positive
N1	HDMI1_CN	AI	HDMI1 Clock - Negative
T2	HDMI1_D2P	AI	HDMI1 Data2 - Positive
T1	HDMI1_D2N	AI	HDMI1 Data2 - Negative
R2	HDMI1_D1P	AI	HDMI1 Data1 - Positive
R1	HDMI1_D1N	AI	HDMI1 Data1 - Negative
P2	HDMI1_D0P	AI	HDMI1 Data0 - Positive
P1	HDMI1_D0N	AI	HDMI1 Data0 - Negative
(*) B1	HDMI2_HPD	B	HDMI2 Hot Plug Detect This signal is driven by the display device to indicate the presence of the display.

Table 5: SupraHD® 748 BGA Signals – HDMI Input

Ball #	Signal / Function	I/O Type	Description
(*) C2	HDMI2_5VSENSE	B	HDMI2 5 Volt Sense Input
(*) C4	HDMI2_SCL	B	HDMI2 Serial Clock This signal is used to assert the data on HDMI_SDA. This signal follows the I2C protocol. This signal should be pulled high with a 47K ohm resistor (to 5V_SENSE on the HDMI connector or to 5V standby power).
(*) C3	HDMI2_SDA	B	HDMI2 Serial Data The HDMI interface sends and receives data on this I2C-compliant signal. This signal should be pulled high with a 47K ohm resistor (to 5V_SENSE on the HDMI connector or to 5V standby power).
J2	HDMI2_CP	AI	HDMI2 Clock - Positive
J1	HDMI2_CN	AI	HDMI2 Clock - Negative
M2	HDMI2_D2P	AI	HDMI2 Data2 - Positive
M1	HDMI2_D2N	AI	HDMI2 Data2 - Negative
L2	HDMI2_D1P	AI	HDMI2 Data1 - Positive
L1	HDMI2_D1N	AI	HDMI2 Data1 - Negative
K2	HDMI2_D0P	AI	HDMI2 Data0 - Positive
K1	HDMI2_D0N	AI	HDMI2 Data0 - Negative
U5	HDMI_REXT	AI	HDMI Current Set This input provides a current source to the HDMI PHY circuitry of the IC. This should be connected to a filtered 3.3V supply through a 390 Ohm 1% resistor.
U3	HDMI_ATEST	AO	HDMI RX Analog Test Test point for Zoran use only.

Table 6: SupraHD® 748 BGA Signals – LVDS & TCON Output

Ball #	Signal / Function	I/O Type	Description
D21	LVDS_CUP	AO	Upper LVDS Channel Clock - Positive
D22	LVDS_CUN	AO	Upper LVDS Channel Clock - Negative
H20	LVDS_D5UP	AO	Upper LVDS Channel D5 - Positive
J20	LVDS_D5UN	AO	Upper LVDS Channel D5 - Negative
J21	LVDS_D4UP	AO	Upper LVDS Channel D4 - Positive
J22	LVDS_D4UN	AO	Upper LVDS Channel D4 - Negative
H21	LVDS_D3UP	AO	Upper LVDS Channel D3 - Positive
H22	LVDS_D3UN	AO	Upper LVDS Channel D3 - Negative
G21	LVDS_D2UP	AO	Upper LVDS Channel D2 - Positive
G22	LVDS_D2UN	AO	Upper LVDS Channel D2 - Negative
F21	LVDS_D1UP	AO	Upper LVDS Channel D1 - Positive
F22	LVDS_D1UN	AO	Upper LVDS Channel D1 - Negative
E21	LVDS_D0UP	AO	Upper LVDS Channel D0 - Positive
E22	LVDS_D0UN	AO	Upper LVDS Channel D0 - Negative
K21	LVDS_CLP	AO	Lower LVDS Channel Clock - Positive
K22	LVDS_CLN	AO	Lower LVDS Channel Clock - Negative

Table 6: SupraHD® 748 BGA Signals – LVDS & TCON Output

Ball #	Signal / Function	I/O Type	Description
N20	LVDS_D5LP	AO	Lower LVDS Channel D5 - Positive
P20	LVDS_D5LN	AO	Lower LVDS Channel D5 - Negative
R21	LVDS_D4LP	AO	Lower LVDS Channel D4 - Positive
R22	LVDS_D4LN	AO	Lower LVDS Channel D4 - Negative
P21	LVDS_D3LP	AO	Lower LVDS Channel D3 - Positive
P22	LVDS_D3LN	AO	Lower LVDS Channel D3 - Negative
N21	LVDS_D2LP	AO	Lower LVDS Channel D2 - Positive
N22	LVDS_D2LN	AO	Lower LVDS Channel D2 - Negative
M21	LVDS_D1LP	AO	Lower LVDS Channel D1 - Positive
M22	LVDS_D1LN	AO	Lower LVDS Channel D1 - Negative
L21	LVDS_D0LP	AO	Lower LVDS Channel D0 - Positive
L22	LVDS_D0LN	AO	Lower LVDS Channel D0 - Negative
Y20	PWM	O	Pulse Width Modulation Output This signal is used to regulate the backlight output of the display panel.
(*) H2	PWM2	B	2nd Pulse Width Modulation Output This signal is used to regulate the backlight output of the display panel.
N18	LVDS_REXT	AI	LVDS External Resistor Connection This should be connected to GND through an 850 Ohm resistor.
L15	LVDS_ATEST	AO	LVDS TX Analog Test Test point for Zoran use only.
W19	TCON_RVS	B	Reverse/Polarity Control Sets the polarity of the source driver data driving. Programmable for the following modes: 1: 1-line/dot inversion 2: 2-line inversion 3: 2-frame inversion + dot inversion 4: 2-frame inversion + 2-line inversion
V22	TCON_TP	B	Transfer/Load Pulse
U21	TCON_STH1	B	Start Pulse Horizontal Specifies the location of the first pixel during loading.
U22	TCON_STH2	B	Start Pulse Horizontal for Send Half Line Specifies the location of the first pixel of the 2nd half of source driver loading if necessary.
V20	TCON_CPV1	B	Clock Pulse Vertical Advance the gate driver to the next row.
W20	TCON_CPV2	B	Clock Pulse Vertical Second CVP.
T22	TCON_OE1	B	Gate Driver/Row Output Enable
V21	TCON_OE2	B	Gate Driver/Row Output Enable Second OE.
T21	TCON_STV1	B	Start Pulse Vertical Specifies the location of the first line during loading.
U20	TCON_STV2	B	Start Pulse Vertical Second STV.
T20	TCON_GPO	B	General Purpose Output Programmable by registers.



Table 7: SupraHD® 748 BGA Signals – Audio Input/Output

Ball #	Signal / Function	I/O Type	Description
C16	AC_L1	AI	Audio Line-level Input #1 - Left
A20	AC_R1	AI	Audio Line-level Input #1 - Right
C17	AC_L2	AI	Audio Line-level Input #2 - Left
B20	AC_R2	AI	Audio Line-level Input #2 - Right
B17	AC_L3	AI	Audio Line-level Input #3 - Left
A19	AC_R3	AI	Audio Line-level Input #3 - Right
A17	AC_L4	AI	Audio Line-level Input #4 - Left
B19	AC_R4	AI	Audio Line-level Input #4 - Right
B18	AC_L5	AI	Audio Line-level Input #5 - Left
A18	AC_R5	AI	Audio Line-level Input #5 - Right
A10	DDX_LP	B	Audio DDX Left Channel Differential Output – Positive The ADATAIO[0] left-channel data is output on this line.
B9	DDX_LN	B	Audio DDX Left Channel Differential Output – Negative The ADATAIO[0] left-channel data is output on this line.
A11	DDX_RP	B	Audio DDX Right Channel Differential Output – Positive The ADATAIO[0] right-channel data is output on this line.
B10	DDX_RN	B	Audio DDX Right Channel Differential Output – Negative The ADATAIO[0] right-channel data is output on this line.
A9	DDX_LO_L	B	Audio DDX Left Channel Line-level Output The ADATAIO[1] left-channel data is output on this line.
C10	DDX_LO_R	B	Audio DDX Right Channel Line-Level Output The ADATAIO[1] right-channel data is output on this line.
(*) A11	DDX_1a	B	Audio DDX Single-Ended Channel Output – ADATAIO[0] Left The ADATAIO[0] left-channel data is output on this line.
(*) B10	DDX_2a	B	Audio DDX Single-Ended Channel Output – ADATAIO[0] Right The ADATAIO[0] right-channel data is output on this line.
(*) A10	DDX_3a	B	Audio DDX Single-Ended Channel Output – ADATAIO[1] Left The ADATAIO[1] left-channel data is output on this line.
(*) B9	DDX_4a	B	Audio DDX Single-Ended Channel Output – ADATAIO[1] Right The ADATAIO[1] right-channel data is output on this line.
(*) A9	DDX_5a	B	Audio DDX Single-Ended Channel Output – ADATAIO[2] Left The ADATAIO[2] left-channel data is output on this line.
(*) C10	DDX_6a	B	Audio DDX Single-Ended Channel Output – ADATAIO[2] Right The ADATAIO[2] right-channel data is output on this line.
C19	IEC958O	O	IEC958 (S/PDIF) Format Out This output carries audio data formatted in IEC 958 format. This signal can be used to drive external hardware audio decoders.
(*) A9, or (*) F1	ACLK	B	Audio Clock This clock is 256 or 384 times the audio sample clock (LRCLK). It is used to generate the IEC-958 formatted output. It is divided to create the BCLK and LRCLK.
(*) A10, or (*) L3	BCLK	B	Bit Clock Software determines whether this pin is an input or an output. As an output, it provides a serial audio clock with software selectable frequency. The clock is typically n times the audio sample rate, where n = 32, 48, 64, or 128. As an input, it represents a serial clock for LRCLK and ADATAO. It is nominally 64 times the audio sample rate. This pin is configured as an input upon device reset.

Table 7: SupraHD® 748 BGA Signals – Audio Input/Output

Ball #	Signal / Function	I/O Type	Description
(*) A11, or (*) F2	LRCLK	B	Left/Right Channel Selector Software determines whether this pin is an input or an output. LRCLK indicates left and right channel selection on the serial audio input; it also indicates the audio sample rate. The polarity of LRCLK is programmable. As an output, LRCLK transitions at the sample rate. As an input, LRCLK determines the sample rate. In this case, the timing accuracy of LRCLK determines the positional accuracy of the output samples.
(*) B9, or (*) G4	ADATAIO[0]	B	Serial Audio Data Output I/O Line 0 This is data line 0 of the I2S audio interface. Data can be either output or input (programmable), clocked by BCLK. All audio channels must have the same sample rate.
(*) C10	ADATAIO[1]	B	Serial Audio Data Output I/O Line 1 This is data line 1 of the I2S audio interface. Data can be either output or input (programmable), clocked by BCLK. All audio channels must have the same sample rate.
(*) B10	ADATAIO[2]	B	Serial Audio Data Output I/O Line 2 This is data line 2 of the I2S audio interface. Data can be either output or input (programmable), clocked by BCLK. All audio channels must have the same sample rate.
G17	AC_VREFP	AO	Reference Voltage for Audio ADC - Positive Internally generated Audio ADC reference voltage pin for off-chip decoupling. Recommended: 10uF cap to AC_VREFN.
G18	AC_VREFN	AO	Reference Voltage for Audio ADC - Negative Internally generated Audio ADC reference voltage pin for off-chip decoupling. Recommended: 10uF cap to AC_VREFP.
F17	AC_RES	AO	Audio ADC External Resistor Reference Recommended: 9.53K (1%) resistor to GND.
F18	AC_VCM	AO	Audio ADC Common Mode Voltage Internally generated audio codec common mode voltage pin for off-chip decoupling. Recommended: 0.1uF cap to GND
F14	AC_LCAP_P	AO	Audio ADC Left Channel Power Filter - Positive Recommended: 220pF cap to AC_LCAP_N.
F15	AC_LCAP_N	AO	Audio ADC Left Channel Power Filter - Negative Recommended: 220pF cap to AC_LCAP_P.
D17	AC_RCAP_P	AO	Audio ADC Right Channel Power Filter - Positive Recommended: 220pF cap to AC_RCAP_N.
D16	AC_RCAP_N	AO	Audio ADC Right Channel Power Filter - Negative Recommended: 220pF cap to AC_RCAP_P.

Table 8: SupraHD® 748 BGA Signals – USB

Ball #	Signal / Function	I/O Type	Description
W22	USB_DP	AI/O	USB 2.0 Data Line - Positive
W21	USB_DN	AI/O	USB 2.0 Data Line - Negative
T18	USB_REXT	AI	USB 2.0 External RSET Signal Input Recommended: 6.04K (1%) pull-down resistor.
V19	USB_ATEST	AO	USB Analog Test Test point for Zoran use only.

Table 9: SupraHD® 748 BGA Signals – DDR2

Ball #	Signal / Function	I/O Type	Description
V17	RDRIVER	AI	RDRIVER Input (Driver Resistor Connection) This resistor is used to calibrate the output drive strength for clock, DQ (output), DQS (output) & DQM (output).
U17	RDRIVER50	AI	RDRIVER50 Input (Driver Resistor Connection) This resistor is used to calibrate the output drive strength for address, command, and CKE.
V16	RTERM	AI	RTERM (Termination Resistor Connection) This resistor is used to calibrate the on-die termination for DQ and DQS inputs.
AB1	S0ADR12	O	DDR2 Address Line 12
Y22	S0ADR11	O	DDR2 Address Line 11
AB4	S0ADR10	O	DDR2 Address Line 10
AA1	S0ADR9	O	DDR2 Address Line 9
AA22	S0ADR8	O	DDR2 Address Line 8
AB2	S0ADR7	O	DDR2 Address Line 7
AB22	S0ADR6	O	DDR2 Address Line 6
AA2	S0ADR5	O	DDR2 Address Line 5
AA21	S0ADR4	O	DDR2 Address Line 4
AB3	S0ADR3	O	DDR2 Address Line 3
AB21	S0ADR2	O	DDR2 Address Line 2
AA3	S0ADR1	O	DDR2 Address Line 1
AA20	S0ADR0	O	DDR2 Address Line 0
AA15	S0DAT15	B	DDR2 Data Line 15
AA9	S0DAT14	B	DDR2 Data Line 14
AA16	S0DAT13	B	DDR2 Data Line 13
AB9	S0DAT12	B	DDR2 Data Line 12
AB11	S0DAT11	B	DDR2 Data Line 11
AB14	S0DAT10	B	DDR2 Data Line 10
AA10	S0DAT9	B	DDR2 Data Line 9
AB16	S0DAT8	B	DDR2 Data Line 8
AA13	S0DAT7	B	DDR2 Data Line 7
AA8	S0DAT6	B	DDR2 Data Line 6
AB12	S0DAT5	B	DDR2 Data Line 5
AA7	S0DAT4	B	DDR2 Data Line 4
AB7	S0DAT3	B	DDR2 Data Line 3
AA12	S0DAT2	B	DDR2 Data Line 2
AB8	S0DAT1	B	DDR2 Data Line 1
AB13	S0DAT0	B	DDR2 Data Line 0
AA17	S0DQS0P	B	DDR2 Data Strobe LDQS
AB17	S0DQS0N	B	DDR2 Data Strobe LDQSN
AA14	S0DQS1P	B	DDR2 Data Strobe UDQS
AB15	S0DQS1N	B	DDR2 Data Strobe UDQSN
AA11	S0DQM0	O	DDR2 Data Mask LDM
AB10	S0DQM1	O	DDR2 Data Mask UDM
AB5	S0BS0	O	DDR2 Bank Select 0

Table 9: SupraHD® 748 BGA Signals – DDR2

Ball #	Signal / Function	I/O Type	Description
AA4	S0BS1	O	DDR2 Bank Select 1
AA5	S0BS2	O	DDR2 Bank Select 2
AA19	S0RASN	O	DDR2 Row Address Strobe
AB20	S0CASN	O	DDR2 Column Address Strobe
AB6	S0WEN	O	DDR2 Write Enable
AB18	S0CLKP	O	DDR2 Clock - Positive
AA18	S0CLKN	B	DDR2 Clock - Negative
AA6	S0CKE	O	DDR2 Clock Enable
V13	S0VREF	AI	DDR2 Voltage Reference May be left unconnected.
AB19	S0ODT	O	On Die Termination

Table 10: SupraHD® 748 BGA Signals – SPI

Ball #	Signal / Function	I/O Type	Description
B21	SPI_CLK	O	SPI Clock
B22	SPI_DI	I	SPI Master Data Input
C22	SPI_DO	B	SPI Master Data Output
C20	SPI_HOLD	B	SPI Master Hold
C21	SPI_SEL0	O	SPI Device Select 0
Y21	SPI_SEL1	B	SPI Device Select 1

Table 11: SupraHD® 748 BGA Signals – UART & I2C

Ball #	Signal / Function	I/O Type	Description
A6	UART0TX	B (open drain)	UART 0 Transmit
C7	UART0RX	B (open drain)	UART 0 Receive
F20	UART1TX	O	UART 1 Transmit
E20	UART1RX	I	UART 1 Receive
(*) A6	TVU_UART_TXD	B (open drain)	TVuC UART Transmit
(*) C7	TVU_UART_RXD	B (open drain)	TVuC UART Receive
F4	I2C0C	B (open drain)	I2C Compatible Clock 0 This signal is the clock of the internal I2C interface between the TVuC and MIPS CPU. This signal can be externally monitored. It is not recommended to use this signal for other than monitoring.
F3	I2C0D	B (open drain)	I2C Compatible Data 0 This signal is the data of the internal I2C interface between the TVuC and MIPS CPU. This signal can be externally monitored. It is not recommended to use this signal for other than monitoring.
A21	I2C1C	B (open drain)	I2C Compatible Clock 1 This signal is the clock of the I2C interface to an external tuner. If this interface to a tuner is not used, it is not recommended to use this signal.
A22	I2C1D	B	I2C Compatible Data 1 This signal is the data of the I2C interface to an external tuner. If this interface to a tuner is not used, it is not recommended to use this signal.



Table 12: SupraHD® 748 BGA Signals – General Purpose I/O

Ball #	Signal / Function	I/O Type	Description
N3	GPIO_P16	B	General Purpose I/O Line 16
J4	GPIO_P14	B	General Purpose I/O Line 14
H1	GPIO_P12	B	General Purpose I/O Line 12
H2	GPIO_P11	B	General Purpose I/O Line 11
H3	GPIO_P10	B	General Purpose I/O Line 10
H4	GPIO_P9	B	General Purpose I/O Line 9
G1	GPIO_P8	B	General Purpose I/O Line 8
G2	GPIO_P7	B	General Purpose I/O Line 7
G3	GPIO_P6	B	General Purpose I/O Line 6
G4	GPIO_P5	B	General Purpose I/O Line 5
F1	GPIO_P4	B	General Purpose I/O Line 4
L3	GPIO_P1	B	General Purpose I/O Line 1
F2	GPIO_P0	B	General Purpose I/O Line 0
(*) Y20	GPIO_S[16]	O	Secondary General Purpose I/O Line 16
(*) C21	GPIO_S[15]	O	Secondary General Purpose I/O Line 15
(*) B21	GPIO_S[14]	O	Secondary General Purpose I/O Line 14
(*) C22	GPIO_S[13]	B	Secondary General Purpose I/O Line 13
(*) C20	GPIO_S[12]	B	Secondary General Purpose I/O Line 12
(*) Y21	GPIO_S[11]	B	Secondary General Purpose I/O Line 11
(*) C19	GPIO_S[10]	O	Secondary General Purpose I/O Line 10
(*) T21	GPIO_S[9]	B	Secondary General Purpose I/O Line 9
(*) T22	GPIO_S[8]	B	Secondary General Purpose I/O Line 8
(*) B22	GPIO_S[7]	I	Secondary General Purpose I/O Line 7
(*) A22	GPIO_S[5]	B	Secondary General Purpose I/O Line 5
(*) A21	GPIO_S[4]	B (open drain)	Secondary General Purpose I/O Line 4
(*) E20	GPIO_S[3]	I	Secondary General Purpose I/O Line 3
(*) F20	GPIO_S[2]	O	Secondary General Purpose I/O Line 2
(*) B2	GPIO_TV_P29	B	TVuC General Purpose I/O Line 29
(*) B5	GPIO_TV_P27	B	TVuC General Purpose I/O Line 27
(*) A5	GPIO_TV_P26	B	TVuC General Purpose I/O Line 26
(*) B6	GPIO_TV_P24	B	TVuC General Purpose I/O Line 24
(*) D6	GPIO_TV_P23	B	TVuC General Purpose I/O Line 23
(*) B4	GPIO_TV_P21	B	TVuC General Purpose I/O Line 21
(*) D5	GPIO_TV_P15	B	TVuC General Purpose I/O Line 15
(*) C6	GPIO_TV_P13	B	TVuC General Purpose I/O Line 13
(*) C5	GPIO_TV_P12	B	TVuC General Purpose I/O Line 12
B1	GPIO_TV_P7	B	TVuC General Purpose I/O Line 7
C2	GPIO_TV_P6	B	TVuC General Purpose I/O Line 6
C3	GPIO_TV_P5	B	TVuC General Purpose I/O Line 5
C4	GPIO_TV_P4	B	TVuC General Purpose I/O Line 4
C1	GPIO_TV_P3	B	TVuC General Purpose I/O Line 3
D2	GPIO_TV_P2	B	TVuC General Purpose I/O Line 2

**Table 12: SupraHD® 748 BGA Signals – General Purpose I/O**

Ball #	Signal / Function	I/O Type	Description
D3	GPIO_TV_P1	B	TVuC General Purpose I/O Line 1
D4	GPIO_TV_P0	B	TVuC General Purpose I/O Line 0

Table 13: SupraHD® 748 BGA Signals – TVuC and Main CPU Signals

Ball #	Signal / Function	I/O Type	Description
E1	RESET_N	I	Reset In Input signal used to reset the TVuC. The TVuC has reset control of the rest of the IC which it can reset as needed via software control.
A2	IRR	I	Infrared Receive
D7	TVCPU_PWM0	O	TVuC PWM
D5	POWER_CTL1	B	Regulator On/Off Control This signal is used to turn regulators on/off during standby mode.
C5	POWER_CTL2	B	Regulator On/Off Control This signal is used to turn regulators on/off during standby mode.
B5	WDT_EN_N	B	Watchdog Timer Enable At boot time, this signal enables or disables the 8051 watchdog timer.
A5	BOOT_OPT/LED	B	Boot From MIPS At boot time, this signal must be pulled low for the MIPS to provide boot code to the 8051. After the boot is completed, this signal can be reconfigured as: LED Driver (Front Panel) This signal is used to drive the LED on the front panel.
(*) F2	BOOT_0	B	Boot Source 0 With BOOT_1, selects the boot source: BOOT[1:0] = 00b, boot from SPI flash. BOOT[1:0] = 01b, boot from UART.
(*) L3	BOOT_1	B	Boot Source 1 With BOOT_0, selects the boot source: BOOT[1:0] = 00b, boot from SPI flash. BOOT[1:0] = 01b, boot from UART.
C6	INT0_N	B	External Interrupt 0 Input used to control timers for pulse width measurement.
D6	INT1_N	B	External Interrupt 1 Input used to control timers for pulse width measurement.
B2	INT2	B	External Interrupt 2 Input used to control timers for pulse width measurement.
B4	T0	B	Timer/Counter 0 External Input
B6	T1	B	Timer/Counter 1 External Input
(*) B5	T2	B	Timer/Counter 2 External Input
D13	ADC8_IN6	AI	ADC8 Input Signal This signal is input to the ADC8 MUX and is designed as a general purpose input for the use of the system designer. This signal must be active while the device is in standby mode.
D9	ADC8_IN3	AI	ADC8 Input Signal This signal is input to the ADC8 MUX and is designed as a general purpose input for the use of the system designer. This signal must be active while the device is in standby mode.

Table 13: SupraHD® 748 BGA Signals – TVuC and Main CPU Signals

Ball #	Signal / Function	I/O Type	Description
C9	ADC8_IN2	AI	ADC8 Input Signal This signal is input to the ADC8 MUX and is designed as a general purpose input for the use of the system designer. This signal must be active while the device is in standby mode.
D8	ADC8_IN1	AI	ADC8 Input Signal This signal is input to the ADC8 MUX and is designed as a general purpose input for the use of the system designer. This signal must be active while the device is in standby mode.
C8	ADC8_IN0	AI	ADC8 Input Signal This signal is input to the ADC8 MUX and is designed as a general purpose input for the use of the system designer. This signal must be active while the device is in standby mode.
F6	ADC8_RBIAIS	AI	ADC8 BIAS Voltage This signal should be pulled to GND via a 10K resistor.

Table 14: SupraHD® 748 BGA Signals – Clocks

Ball #	Signal / Function	I/O Type	Description
B8	CLKIN_25M	AI	External Crystal Input An external 25 MHz crystal is connected to CLKIN_25M & CLKOUT_25M to provide the system clock. Typically CLKOUT_25M is buffered and sent to CLKIN_25M, which provides the clock reference to the digital portion of the IC. Crystal Specification: 50ppm 0-70 degrees C
A8	CLKOUT_25M	AO	External Crystal Output An external 25 MHz crystal is connected to CLKIN_25M & CLKOUT_25M to provide the system clock. Typically CLKOUT_25M is buffered and sent to CLKIN_25M, which provides the clock reference to the digital portion of the IC. Crystal Specification: 50ppm 0-70 degrees C
A7	CLKIN_RTC	AI	Real-Time Clock (RTC) External Crystal (Optional) An external 32.768 KHz crystal is connected to CLKIN_RTC and CLKOUT_RTC to provide the RTC to the TVuC. If a crystal is not used, an internal RC oscillator circuit can be used instead. Crystal Specification: 50ppm 0-70 degrees C
B7	CLKOUT_RTC	AO	Real-Time Clock (RTC) External Crystal (Optional) An external 32.768 KHz crystal is connected to CLKIN_RTC and CLKOUT_RTC to provide the RTC to the TVuC. If a crystal is not used, an internal RC oscillator circuit can be used instead. Crystal Specification: 50ppm 0-70 degrees C

Table 15: SupraHD® 748 BGA Signals – Power and Ground

Ball #	Signal / Function	I/O Type	Description
H17	CORE_VDD	P	Core Voltage Power Supply (1.1V)
J6	CORE_VDD	P	Core Voltage Power Supply (1.1V)
R5	CORE_VDD	P	Core Voltage Power Supply (1.1V)
R17	CORE_VDD	P	Core Voltage Power Supply (1.1V)
R18	CORE_VDD	P	Core Voltage Power Supply (1.1V)
T5	CORE_VDD	P	Core Voltage Power Supply (1.1V)
T6	CORE_VDD	P	Core Voltage Power Supply (1.1V)
G7	COREVDD_STBY	P	Core Voltage Standby Power Supply (1.1V)
U9	MVDD	P	Memory Interface Power Supply (1.8V)
U13	MVDD	P	Memory Interface Power Supply (1.8V)
U14	MVDD	P	Memory Interface Power Supply (1.8V)
V8	MVDD	P	Memory Interface Power Supply (1.8V)
V9	MVDD	P	Memory Interface Power Supply (1.8V)
V14	MVDD	P	Memory Interface Power Supply (1.8V)
K17	IOVDD	P	IO Voltage Power Supply (3.3V)
K18	IOVDD	P	IO Voltage Power Supply (3.3V)
L5	IOVDD	P	IO Voltage Power Supply (3.3V)
L6	IOVDD	P	IO Voltage Power Supply (3.3V)
G6	IOVDD_STBY	P	IO Voltage Standby Power Supply (3.3V)
H6	IOVDD_STBY	P	IO Voltage Standby Power Supply (3.3V)
U11	CVDD_MSD	P	Memory Interface Power Supply (1.1V)
V11	CVDD_MSD	P	Memory Interface Power Supply (1.1V)
R10	CVSS_MSD	P	Memory Interface Ground (for 1.1V Supply)
R11	CVSS_MSD	P	Memory Interface Ground (for 1.1V Supply)
R12	CVSS_MSD	P	Memory Interface Ground (for 1.1V Supply)
R13	CVSS_MSD	P	Memory Interface Ground (for 1.1V Supply)
N5	HDMI_VDD3P3	P	HDMI Interface Power Supply (3.3V)
N6	HDMI_VDD3P3	P	HDMI Interface Power Supply (3.3V)
R8	HDMI_VDD3P3	P	HDMI Interface Power Supply (3.3V)
R6	HDMI_VDD1V	P	HDMI Interface Power Supply (1.1V)
L17	LDI_VDD	P	LVDS Power Supply (3.3V)
L18	LDI_VDD	P	LVDS Power Supply (3.3V)
N17	LDI_CVDD	P	LVDS TX Power Supply (1.1V)
T17	USB2_VDDA33T	P	USB 2.0 PHY Power Supply (3.3V)
U16	DDRPLL_AVDD	P	DDR2 PLL Power Supply (3.3V) Connect to 3.3V through a 1.5Kohm resistor.
R15	DDRPLL_GND	P	DDR2 PLL Ground Recommended: 10nF cap to DDRPLL_AVDD. Do not connect to GND on PCB.
G8	VDDIO_SYSPLL	P	On-Chip PLL Power Supply (3.3V) Connect to 3.3V through a 1.5Kohm resistor.
H10	GNDSYSPLL	P	On-Chip PLL Ground Recommended: 10nF cap to VDDIO_SYSPLL. Do not connect to GND on PCB.
F7	CLK25M_VDD	P	Crystal Circuit Power Supply (3.3V)



SupraHD®-748-DTS-GEN-v1.1

Table 15: SupraHD® 748 BGA Signals – Power and Ground

Ball #	Signal / Function	I/O Type	Description
F16	AC_AVDD	P	Audio ADC Power Supply (3.3V)
H15	AC_AGNDREF	P	Audio ADC Reference Ground
F12	AFE_VPWR33	P	Analog Front End Power Supply (3.3V)
F11	AFE_VPWR33_REF	P	AFE REF Power Supply (3.3V)
F13	AFE_VPWR10	P	Analog Front End Power Supply (1.1V)
H12	AFE_GND	P	Analog Front End Ground
D11	AFE_VPWR33_PLL	P	AFE PLL Power Supply (3.3V)
F10	AFE_VPWR10_PLL	P	AFE PLL Power Supply (1.1V)
H11	AFE_PLL_GND	P	AFE PLL Ground
F8	DDX_VDDIO	P	DDX Audio Power Supply (3.3V)
F9	AUDPLL_VDDIO	P	Audio PLL Power Supply (3.3V)
H9	AUDPLL_GND	P	Audio PLL Ground
G20	GND	P	Ground
H13	GND	P	Ground
H14	GND	P	Ground
J8	GND	P	Ground
J9	GND	P	Ground
J10	GND	P	Ground
J11	GND	P	Ground
J12	GND	P	Ground
J13	GND	P	Ground
J14	GND	P	Ground
J15	GND	P	Ground
K3	GND	P	Ground
K8	GND	P	Ground
K9	GND	P	Ground
K10	GND	P	Ground
K11	GND	P	Ground
K12	GND	P	Ground
K13	GND	P	Ground
K14	GND	P	Ground
K15	GND	P	Ground
K20	GND	P	Ground
L8	GND	P	Ground
L9	GND	P	Ground
L10	GND	P	Ground
L11	GND	P	Ground
L12	GND	P	Ground
L13	GND	P	Ground
L14	GND	P	Ground
L20	GND	P	Ground
M3	GND	P	Ground
M8	GND	P	Ground



SupraHD®-748-DTS-GEN-v1.1

Table 15: SupraHD® 748 BGA Signals – Power and Ground

Ball #	Signal / Function	I/O Type	Description
M9	GND	P	Ground
M10	GND	P	Ground
M11	GND	P	Ground
M12	GND	P	Ground
M13	GND	P	Ground
M14	GND	P	Ground
M15	GND	P	Ground
M20	GND	P	Ground
N8	GND	P	Ground
N9	GND	P	Ground
N10	GND	P	Ground
N11	GND	P	Ground
N12	GND	P	Ground
N13	GND	P	Ground
N14	GND	P	Ground
N15	GND	P	Ground
P3	GND	P	Ground
P8	GND	P	Ground
P9	GND	P	Ground
P10	GND	P	Ground
P11	GND	P	Ground
P12	GND	P	Ground
P13	GND	P	Ground
P14	GND	P	Ground
P15	GND	P	Ground
R3	GND	P	Ground
R9	GND	P	Ground
R14	GND	P	Ground
R20	GND	P	Ground
T3	GND	P	Ground
V3	GND	P	Ground
W3	GND	P	Ground
Y3	GND	P	Ground
Y4	GND	P	Ground
Y5	GND	P	Ground
Y6	GND	P	Ground
Y7	GND	P	Ground
Y8	GND	P	Ground
Y9	GND	P	Ground
Y10	GND	P	Ground
Y11	GND	P	Ground
Y12	GND	P	Ground
Y13	GND	P	Ground



SupraHD®-748-DTS-GEN-v1.1

Table 15: SupraHD® 748 BGA Signals – Power and Ground

Ball #	Signal / Function	I/O Type	Description
Y14	GND	P	Ground
Y15	GND	P	Ground
Y16	GND	P	Ground
Y17	GND	P	Ground
Y18	GND	P	Ground
Y19	GND	P	Ground

2.2. QFP Package Signal Lists

The following tables list the signals in numeric pin order, by signals that have one or more alternate functions and by functional group.

2.2.1. Signal Listing by Pin Order

The following table lists the QFP package signals in pin order. The following information is provided for each signal:

- Pin number
- Signal name - primary function for signals with multiple function options
- I/O type
 - I = digital input, O = digital output, B = digital input/output
 - AI = analog input, AO = analog output, AI/O = analog input/output, P = power/ground
- Indication if the signal includes an internal pull-up or pull-down resistor
 - P.U. = pull-up, P.D. = pull-down, x = not applicable
- Default state of the signal immediately after RESET (after /RESET_n is pulled high)
 - I = input, O = output

Table 16: SupraHD® 748 QFP Signals – Pin Order

Pin #	Signal Name	I/O Type	Internal pull-up / down?	Default after RESET
1	ADC8_IN2	AI	x	x
2	ADC8_IN1	AI	x	x
3	ADC8_IN0	AI	x	x
4	COREVDD_STBY	P	x	x
5	ADC8_RBIAS	AI	x	x
6	IOVDD_STBY	P	x	x
7	TVCPU_PWM0	O	x	Low
8	UART0RX	B (open drain)	x	I
9	UART0TX	B (open drain)	x	O (High)
10	INT0_N	B	x	I
11	INT1_N	B	x	I
12	BOOT_OPT/LED	B	x	I
13	WDT_EN_N	B	x	I
14	POWER_CTL2	B	x	O (High)
15	POWER_CTL1	B	x	O (High)
16	IF_AGC	O (open drain)	x	Low
17	COREVDD_STBY	P	x	x
18	VGA_SDA	B (open drain)	x	I
19	VGA_SCL	B (open drain)	x	I
20	IRR	I	x	I
21	INT2	B	x	I
22	HDMI_CEC	B (open drain)	x	I
23	GPIO_TV_P4	B	x	I
24	GPIO_TV_P5	B	x	I
25	GPIO_TV_P6	B	x	I

Table 16: SupraHD® 748 QFP Signals – Pin Order

Pin #	Signal Name	I/O Type	Internal pull-up / down?	Default after RESET
26	GPIO_TV_P7	B	x	l
27	GPIO_TV_P0	B	x	l
28	GPIO_TV_P1	B	x	l
29	GPIO_TV_P2	B	x	l
30	GPIO_TV_P3	B	x	l
31	HDMI0_SCL	B (open drain)	x	l
32	HDMI0_SDA	B (open drain)	x	l
33	HDMI0_HPD	B	x	l
34	HDMI0_5VSNS	B	x	l
35	RESET_N	I	x	l
36	IOVDD_STBY	P	x	x
37	CORE_VDD	P	x	x
38	I2C0C	B (open drain)	x	l
39	I2C0D	B (open drain)	x	l
40	GPIO_P0	B	PD	l
41	GPIO_P1	B	PU	l
42	IOVDD	P	x	x
43	GPIO_P6	B	PU	l
44	GPIO_P7	B	PU	l
45	GPIO_P8	B	PU	l
46	IOVDD	P	x	x
47	GPIO_P9	B	PU	l
48	GPIO_P10	B	PU	l
49	IOVDD	P	x	x
50	AFE_HSYNC_IN	B	x	l
51	CORE_VDD	P	x	x
52	GND	P	x	x
53	HDMI_VDD3P3	P	x	x
54	HDMI_VDD1V	P	x	x
55	HDMI_VDD1V	P	x	x
56	HDMI1_CN	AI	x	x
57	HDMI1_CP	AI	x	x
58	HDMI1_D0N	AI	x	x
59	HDMI1_D0P	AI	x	x
60	HDMI1_D1N	AI	x	x
61	HDMI1_D1P	AI	x	x
62	HDMI1_D2N	AI	x	x
63	HDMI1_D2P	AI	x	x
64	HDMI_VDD3P3	P	x	x
65	HDMI_REXT	AI	x	x
66	HDMI_VDD3P3	P	x	x
67	HDMI_VDD1V	P	x	x
68	HDMI_VDD1V	P	x	x

Table 16: SupraHD® 748 QFP Signals – Pin Order

Pin #	Signal Name	I/O Type	Internal pull-up/down?	Default after RESET
69	CORE_VDD	P	x	x
70	HDMI_VDD3P3	P	x	x
71	MVDD	P	x	x
72	S0ADR9	O	x	Low
73	S0ADR12	O	x	Low
74	S0ADR7	O	x	Low
75	CORE_VDD	P	x	x
76	S0ADR5	O	x	Low
77	S0ADR3	O	x	Low
78	S0ADR1	O	x	Low
79	S0ADR10	O	x	Low
80	S0BS1	O	x	Low
81	CVDD_MSD	P	x	x
82	S0BS2	O	x	Low
83	S0BS0	O	x	Low
84	S0CKE	O	x	Low
85	S0WEN	O	x	Low
86	CORE_VDD	P	x	x
87	S0DAT4	B	x	I
88	S0DAT3	B	x	I
89	S0DAT1	B	x	I
90	S0DAT6	B	x	I
91	S0DAT12	B	x	I
92	S0DAT11	B	x	I
93	CVDD_MSD	P	x	x
94	S0DAT14	B	x	I
95	S0DAT9	B	x	I
96	S0DQM1	O	x	Low
97	S0DQM0	O	x	Low
98	S0DAT5	B	x	I
99	S0DAT2	B	x	I
100	S0DAT0	B	x	I
101	MVDD	P	x	x
102	S0DAT7	B	x	I
103	S0DQS1P	B	x	I
104	S0DQS1N	B	x	I
105	S0DAT8	B	x	I
106	S0DAT15	B	x	I
107	S0DAT10	B	x	I
108	S0DAT13	B	x	I
109	S0DQS0P	B	x	I
110	S0DQS0N	B	x	I
111	CVDD_MSD	P	x	x

Table 16: SupraHD® 748 QFP Signals – Pin Order

Pin #	Signal Name	I/O Type	Internal pull-up/down?	Default after RESET
112	S0CLKP	O	x	High
113	S0CLKN	B	x	I
114	CORE_VDD	P	x	x
115	S0ODT	O	x	Low
116	S0RASN	O	x	Low
117	S0CASN	O	x	Low
118	S0ADR0	O	x	Low
119	S0ADR2	O	x	Low
120	S0ADR4	O	x	Low
121	S0ADR6	O	x	Low
122	CVDD_MSD	P	x	x
123	S0ADR8	O	x	Low
124	S0ADR11	O	x	Low
125	CORE_VDD	P	x	x
126	DDRPLL_GND	P	x	x
127	DDRPLL_AVDD	P	x	x
128	MVDD	P	x	x
129	CORE_VDD	P	x	x
130	USB_DN	AI/O	x	x
131	USB2_VDDA33T	P	x	x
132	USB_DP	AI/O	x	x
133	USB_REXT	AI	x	x
134	PWM	O	x	Low
135	TCON_RVS	B	PU	I
136	TCON_CPV2	B	PU	I
137	IOVDD	P	x	x
138	TCON_TP	B	PU	I
139	TCON_OE2	B	PU	I
140	TCON_CPV1	B	PU	I
141	IOVDD	P	x	x
142	TCON_STV2	B	PU	I
143	TCON_OE1	B	PU	I
144	TCON_STV1	B	PU	I
145	TCON_GPO	B	PU	I
146	LDI_VDD	P	x	x
147	LDI_CVDD	P	x	x
148	LVDS_D5LP	AO	x	x
149	LVDS_D5LN	AO	x	x
150	LVDS_D4LP	AO	x	x
151	LVDS_D4LN	AO	x	x
152	LVDS_D3LP	AO	x	x
153	LVDS_D3LN	AO	x	x
154	LVDS_D2LP	AO	x	x

Table 16: SupraHD® 748 QFP Signals – Pin Order

Pin #	Signal Name	I/O Type	Internal pull-up / down?	Default after RESET
155	LVDS_D2LN	AO	x	x
156	LVDS_D1LP	AO	x	x
157	LVDS_D1LN	AO	x	x
158	LVDS_D0LP	AO	x	x
159	LVDS_D0LN	AO	x	x
160	LVDS_CLP	AO	x	x
161	LVDS_CLN	AO	x	x
162	LDI_VDD	P	x	x
163	LVDS_REXT	AI	x	x
164	LVDS_D5UP	AO	x	x
165	LVDS_D5UN	AO	x	x
166	LVDS_D4UP	AO	x	x
167	LVDS_D4UN	AO	x	x
168	LVDS_D3UP	AO	x	x
169	LVDS_D3UN	AO	x	x
170	LVDS_D2UP	AO	x	x
171	LVDS_D2UN	AO	x	x
172	LVDS_D1UP	AO	x	x
173	LVDS_D1UN	AO	x	x
174	LVDS_D0UP	AO	x	x
175	LVDS_D0UN	AO	x	x
176	LVDS_CUP	AO	x	x
177	LVDS_CUN	AO	x	x
178	LDI_CVDD	P	x	x
179	LDI_VDD	P	x	x
180	CORE_VDD	P	x	x
181	IOVDD	P	x	x
182	TCK	B	PU	I
183	TDI	B	PU	I
184	TDO	B	x	I
185	TMS	B	PU	I
186	IEC958O	O	x	O (Low)
187	SPI_DO	B	x	O (Low)
188	SPI_DI	I	x	I
189	SPI_HOLD	B	PU	O (High)
190	SPI_SEL0	O	x	High
191	SPI_CLK	O	x	Low
192	IOVDD	P	x	x
193	I2C1C	B (open drain)	x	I
194	I2C1D	B	x	I
195	AC_R1	AI	x	x
196	AC_R2	AI	x	x
197	AC_R3	AI	x	x

Table 16: SupraHD® 748 QFP Signals – Pin Order

Pin #	Signal Name	I/O Type	Internal pull-up/down?	Default after RESET
198	AC_R4	AI	x	x
199	AC_RCAP_P	AO	x	x
200	AC_RCAP_N	AO	x	x
201	AC_AVDD	P	x	x
202	AC_AGNDREF	P	x	x
203	AC_RES	AO	x	x
204	AC_VREFP	AO	x	x
205	AC_VREFN	AO	x	x
206	AC_LCAP_N	AO	x	x
207	AC_LCAP_P	AO	x	x
208	AC_L4	AI	x	x
209	AC_L3	AI	x	x
210	AC_L2	AI	x	x
211	AC_L1	AI	x	x
212	IOVDD	P	x	x
213	CORE_VDD	P	x	x
214	AFE_VGA_B	AI	x	x
215	AFE_SVID_C	AI	x	x
216	AFE_PB	AI	x	x
217	AFE_VPWR33	P	x	x
218	AFE_GND	P	x	x
219	AFE_VPWR10	P	x	x
220	AFE_VGA_G	AI	x	x
221	AFE_SVID_Y	AI	x	x
222	AFE_CVBS1	AI	x	x
223	AFE_Y	AI	x	x
224	AFE_VPWR33	P	x	x
225	AFE_GND	P	x	x
226	AFE_VPWR10	P	x	x
227	AFE_VGA_R	AI	x	x
228	AFE_IFN	AI	x	x
229	AFE_IFP	AI	x	x
230	AFE_PR	AI	x	x
231	AFE_VPWR33	P	x	x
232	AFE_GND	P	x	x
233	AFE_VPWR10	P	x	x
234	AFE_RSET	AI	x	x
235	AFE_VPWR33_REF	P	x	x
236	AFE_VPWR33_PLL	P	x	x
237	AFE_PLL_GND	P	x	x
238	AFE_FILTOUT	AO	x	x
239	AFE_FILTOUT2	AO	x	x
240	AFE_VPWR10_PLL	P	x	x

Table 16: SupraHD® 748 QFP Signals – Pin Order

Pin #	Signal Name	I/O Type	Internal pull-up / down?	Default after RESET
241	AFE_SOG	AI	x	x
242	AFE_SOY	AI	x	x
243	VDDIO_SYSPLL	P	x	x
244	AUDPLL_VDDIO	P	x	x
245	AUDPLL_GND	P	x	x
246	DDX_RP	B	PU	O (Low)
247	DDX_RN	B	PU	O (Low)
248	DDX_VDDIO	P	x	x
249	DDX_LP	B	PU	O (Low)
250	DDX_LN	B	PU	O (Low)
251	DDX_LO_L	B	PU	O (Low)
252	DDX_LO_R	B	PU	O (Low)
253	CLKIN_25M	AI	x	x
254	CLKOUT_25M	AO	x	x
255	CLK25M_VDD	P	x	x
256	CORE_VDD	P	x	x
e-pad	GND	P	x	x

2.2.2. Signals with Alternate Functions

The following table lists the QFP package signals that offer alternate functionality. The signals are listed in pin order. Signals not listed do not have any alternate functionality. The following information is provided for each signal:

- Pin number
- Signal name / 1st function – this is the primary function of the signal
- 2nd function – this is the secondary function of the signal
- 3rd function – this is the tertiary function of the signal
- 4th function – this is the fourth function of the signal

Function columns with blank entries in the 3rd and/or 4th function columns indicate that additional functionality is not applicable.

Table 17: SupraHD® 748 QFP Signals with Alternate Functions

Pin #	Signal Name / 1 st Function	2 nd Function	3 rd Function	4 th Function
8	UART0RX	TVU_UART_RXD		
9	UART0TX	TVU_UART_TXD		
10	INT0_N	GPIO_TV_P13	Reserved for debug	
11	INT1_N	GPIO_TV_P23	Reserved for debug	
12	BOOT_OPT/LED	GPIO_TV_P26	Reserved for debug	
13	WDT_EN_N	GPIO_TV_P27	T2	Reserved for debug
14	POWER_CTL2	GPIO_TV_P12	Reserved for debug	
15	POWER_CTL1	GPIO_TV_P15	Reserved for debug	
18	VGA_SDA	Reserved for debug		
19	VGA_SCL	Reserved for debug		
21	INT2	AFE_VSYNC_IN	GPIO_TV_P29	Reserved for debug
23	GPIO_TV_P4	HDMI2_SCL		
24	GPIO_TV_P5	HDMI2_SDA		
25	GPIO_TV_P6	HDMI2_5VSENSE		
26	GPIO_TV_P7	HDMI2_HPD		
27	GPIO_TV_P0	HDMI1_SCL		
28	GPIO_TV_P1	HDMI1_SDA		
29	GPIO_TV_P2	HDMI1_HPD		
30	GPIO_TV_P3	HDMI1_5VSENSE		
40	GPIO_P0	BOOT_0	LRCLK	Reserved for debug
41	GPIO_P1	BOOT_1	BCLK	Reserved for debug
43	GPIO_P6	Reserved for debug		
44	GPIO_P7	Reserved for debug		
45	GPIO_P8	Reserved for debug		
47	GPIO_P9	Reserved for debug		
48	GPIO_P10	Reserved for debug		
50	AFE_HSYNC_IN	Reserved for debug		
134	PWM	GPIO_S[16]		
135	TCON_RVS	Reserved for debug	Reserved for debug	
136	TCON_CPV2	TAPSEL_CAS	Reserved for debug	

Table 17: SupraHD® 748 QFP Signals with Alternate Functions

Pin #	Signal Name / 1 st Function	2 nd Function	3 rd Function	4 th Function
138	TCON_TP	TAPSEL	Reserved for debug	
139	TCON_OE2	Reserved for debug		
140	TCON_CPV1	Reserved for debug		
142	TCON_STV2	Reserved for debug		
143	TCON_OE1	GPIO_S[8]		
144	TCON_STV1	GPIO_S[9]		
145	TCON_GPO	Reserved for debug		
182	TCK	EJTCK	Reserved for debug	
183	TDI	EJTDI	Reserved for debug	
184	TDO	EJTDO	Reserved for debug	
185	TMS	EJTMS	Reserved for debug	
186	IEC958O	GPIO_S[10]	Reserved for debug	
187	SPI_DO	GPIO_S[13]		
188	SPI_DI	GPIO_S[7]		
189	SPI_HOLD	GPIO_S[12]		
190	SPI_SELO	GPIO_S[15]		
191	SPI_CLK	GPIO_S[14]		
193	I2C1C	GPIO_S[4]		
194	I2C1D	GPIO_S[5]		
246	DDX_RP	LRCLK	DDX_1a	Reserved for debug
247	DDX_RN	ADATAIO[2]	DDX_2a	Reserved for debug
249	DDX_LP	BCLK	DDX_3a	Reserved for debug
250	DDX_LN	ADATAIO[0]	DDX_4a	Reserved for debug
251	DDX_LO_L	ACLK	DDX_5a	Reserved for debug
252	DDX_LO_R	ADATAIO[1]	DDX_6a	Reserved for debug

2.2.3. Signal Listings by Function

The following tables list the QFP package signals grouped according to function. The following information is provided for each signal:

- Pin number - numbers with a preceding (*) indicate 2nd / 3rd / 4th functionality for the signal
- Signal/function name - can be the primary, secondary, tertiary or 4th function of the signal
- I/O type
 - I = digital input, O = digital output, B = digital input/output
 - AI = analog input, AO = analog output, AI/O = analog input/output, P = power/ground
- Function description

Table 18: SupraHD® 748 QFP Signals – JTAG/EJTAG

Pin #	Signal / Function	I/O Type	Description
(*) 138	TAPSEL	B	EJTAG Select This signal determines the functionality of the JTAG/EJTAG signals: When low, JTAG functionality is enabled. When high, EJTAG functionality is enabled. Default is EJTAG.
182	TCK	B	JTAG Clock
183	TDI	B	JTAG Data In
184	TDO	B	JTAG Data Out
185	TMS	B	JTAG Mode Select
(*) 136	TAPSEL_CAS	B	Test Signal
(*) 182	EJTCK	B	EJTAG Clock
(*) 183	EJTDI	B	EJTAG Data In
(*) 184	EJTDO	B	EJTAG Data Out
(*) 185	EJTMS	B	EJTAG Mode Select

Table 19: SupraHD® 748 QFP Signals – Analog Video Front-End Input

Pin #	Signal / Function	I/O Type	Description
229	AFE_IFP	AI	Analog IF positive Differential input. Analog input to Main ADC. This input has built-in ESD protection rated up to 2000V. Input resistance - 25 KOhm
228	AFE_IFN	AI	Analog IF Negative Differential input. Analog input to Main ADC. This input has built-in ESD protection rated up to 2000V. Input resistance - 25 KOhm
16	IF_AGC	O (open drain)	IF AGC Signal
222	AFE_CVBS1	AI	Composite Video Input 1 This input has built-in ESD protection rated up to 2000V.
221	AFE_SVID_Y	AI	S-Video Y (Luma) Input This input has built-in ESD protection rated up to 2000V.
215	AFE_SVID_C	AI	S-Video C (Chroma) Input This input has built-in ESD protection rated up to 2000V.
223	AFE_Y	AI	Component Y Input This input has built-in ESD protection rated up to 2000V.

Table 19: SupraHD® 748 QFP Signals – Analog Video Front-End Input

Pin #	Signal / Function	I/O Type	Description
216	AFE_PB	AI	Component Pb Input This input has built-in ESD protection rated up to 2000V.
230	AFE_PR	AI	Component Pr Input This input has built-in ESD protection rated up to 2000V.
242	AFE_SOY	AI	Sync-On-Y (Luma) Input
227	AFE_VGA_R	AI	VGA Red Input This input has built-in ESD protection rated up to 2000V.
220	AFE_VGA_G	AI	VGA Green Input This input has built-in ESD protection rated up to 2000V.
214	AFE_VGA_B	AI	VGA Blue Input This input has built-in ESD protection rated up to 2000V.
241	AFE_SOG	AI	Sync-On-Green Input
50	AFE_HSYNC_IN	B	Horizontal Sync Input for VGA
(*) 21	AFE_VSYNC_IN	B	Vertical Sync Input for VGA
19	VGA_SCL	B (open drain)	VGA I2C Clock I2C clock for DDC information. This signal should be pulled high with a 47K ohm resistor (to 5V_VGA on the VGA connector or to 5V standby power).
18	VGA_SDA	B (open drain)	VGA I2C Data I2C data for DDC information This signal should be pulled high with a 47K ohm resistor (to 5V_VGA on the VGA connector or to 5V standby power).
238	AFE_FILTOUT	AO	AFE PLL Filter Filter connection to R/C filter. Recommended: 200pF cap to AFE 1.1V.
239	AFE_FILTOUT2	AO	AFE PLL Filter 2 Filter connection to R/C filter. This signal includes an internal, programmable series resistor. Recommended: 3.9nF cap to AFE 1.1V
234	AFE_RSET	AI	ADC Current Reference Setting This signal provides a current reference to the video ADCs. This signal has built-in ESD protection rated up to 2000V. Recommended: 0.1uF cap to GND.

Table 20: SupraHD® 748 QFP Signals – HDMI Input

Pin #	Signal / Function	I/O Type	Description
22	HDMI_CEC	B (open drain)	HDMI Consumer Electronics Control
33	HDMI0_HPD	B	HDMI0 Hot Plug Detect This signal is driven by the display device to indicate the presence of the display.
34	HDMI0_5VSNS	B	HDMI0 5 Volt Sense Input
31	HDMI0_SCL	B (open drain)	HDMI0 Serial Clock This signal is used to assert the data on HDMI_SDA. This signal follows the I2C protocol. This signal should be pulled high with a 47K ohm resistor (to 5V_SENSE on the HDMI connector or to 5V standby power).

Table 20: SupraHD® 748 QFP Signals – HDMI Input

Pin #	Signal / Function	I/O Type	Description
32	HDMI0_SDA	B (open drain)	HDMI0 Serial Data The HDMI interface sends and receives data on this I2C-compliant signal. This signal should be pulled high with a 47K ohm resistor (to 5V_SENSE on the HDMI connector or to 5V standby power).
(*) 29	HDMI1_HPD	B	HDMI1 Hot Plug Detect This signal is driven by the display device to indicate the presence of the display.
(*) 30	HDMI1_5VSENSE	B	HDMI1 5 Volt Sense Input
(*) 27	HDMI1_SCL	B	HDMI1 Serial Clock This signal is used to assert the data on HDMI_SDA. This signal follows the I2C protocol. This signal should be pulled high with a 47K ohm resistor (to 5V_SENSE on the HDMI connector or to 5V standby power).
(*) 28	HDMI1_SDA	B	HDMI1 Serial Data The HDMI interface sends and receives data on this I2C-compliant signal. This signal should be pulled high with a 47K ohm resistor (to 5V_SENSE on the HDMI connector or to 5V standby power).
57	HDMI1_CP	AI	HDMI1 Clock - Positive
56	HDMI1_CN	AI	HDMI1 Clock - Negative
63	HDMI1_D2P	AI	HDMI1 Data2 - Positive
62	HDMI1_D2N	AI	HDMI1 Data2 - Negative
61	HDMI1_D1P	AI	HDMI1 Data1 - Positive
60	HDMI1_D1N	AI	HDMI1 Data1 - Negative
59	HDMI1_D0P	AI	HDMI1 Data0 - Positive
58	HDMI1_D0N	AI	HDMI1 Data0 - Negative
(*) 26	HDMI2_HPD	B	HDMI2 Hot Plug Detect This signal is driven by the display device to indicate the presence of the display.
(*) 25	HDMI2_5VSENSE	B	HDMI2 5 Volt Sense Input
(*) 23	HDMI2_SCL	B	HDMI2 Serial Clock This signal is used to assert the data on HDMI_SDA. This signal follows the I2C protocol. This signal should be pulled high with a 47K ohm resistor (to 5V_SENSE on the HDMI connector or to 5V standby power).
(*) 24	HDMI2_SDA	B	HDMI2 Serial Data The HDMI interface sends and receives data on this I2C-compliant signal. This signal should be pulled high with a 47K ohm resistor (to 5V_SENSE on the HDMI connector or to 5V standby power).
65	HDMI_REXT	AI	HDMI Current Set This input provides a current source to the HDMI PHY circuitry of the IC. This should be connected to a filtered 3.3V supply through a 390 Ohm 1% resistor.

Table 21: SupraHD® 748 QFP Signals – LVDS & TCON Output

Pin #	Signal / Function	I/O Type	Description
176	LVDS_CUP	AO	Upper LVDS Channel Clock - Positive
177	LVDS_CUN	AO	Upper LVDS Channel Clock - Negative
164	LVDS_D5UP	AO	Upper LVDS Channel D5 - Positive
165	LVDS_D5UN	AO	Upper LVDS Channel D5 - Negative
166	LVDS_D4UP	AO	Upper LVDS Channel D4 - Positive
167	LVDS_D4UN	AO	Upper LVDS Channel D4 - Negative
168	LVDS_D3UP	AO	Upper LVDS Channel D3 - Positive
169	LVDS_D3UN	AO	Upper LVDS Channel D3 - Negative
170	LVDS_D2UP	AO	Upper LVDS Channel D2 - Positive
171	LVDS_D2UN	AO	Upper LVDS Channel D2 - Negative
172	LVDS_D1UP	AO	Upper LVDS Channel D1 - Positive
173	LVDS_D1UN	AO	Upper LVDS Channel D1 - Negative
174	LVDS_D0UP	AO	Upper LVDS Channel D0 - Positive
175	LVDS_D0UN	AO	Upper LVDS Channel D0 - Negative
160	LVDS_CLP	AO	Lower LVDS Channel Clock - Positive
161	LVDS_CLN	AO	Lower LVDS Channel Clock - Negative
148	LVDS_D5LP	AO	Lower LVDS Channel D5 - Positive
149	LVDS_D5LN	AO	Lower LVDS Channel D5 - Negative
150	LVDS_D4LP	AO	Lower LVDS Channel D4 - Positive
151	LVDS_D4LN	AO	Lower LVDS Channel D4 - Negative
152	LVDS_D3LP	AO	Lower LVDS Channel D3 - Positive
153	LVDS_D3LN	AO	Lower LVDS Channel D3 - Negative
154	LVDS_D2LP	AO	Lower LVDS Channel D2 - Positive
155	LVDS_D2LN	AO	Lower LVDS Channel D2 - Negative
156	LVDS_D1LP	AO	Lower LVDS Channel D1 - Positive
157	LVDS_D1LN	AO	Lower LVDS Channel D1 - Negative
158	LVDS_D0LP	AO	Lower LVDS Channel D0 - Positive
159	LVDS_D0LN	AO	Lower LVDS Channel D0 - Negative
134	PWM	O	Pulse Width Modulation Output This signal is used to regulate the backlight output of the display panel.
163	LVDS_REXT	AI	LVDS External Resistor Connection This should be connected to GND through an 850 Ohm resistor.
135	TCON_RVS	B	Reverse/Polarity Control Sets the polarity of the source driver data driving. Programmable for the following modes: 1: 1-line/dot inversion 2: 2-line inversion 3: 2-frame inversion + dot inversion 4: 2-frame inversion + 2-line inversion
138	TCON_TP	B	Transfer/Load Pulse
140	TCON_CPV1	B	Clock Pulse Vertical Advance the gate driver to the next row.
136	TCON_CPV2	B	Clock Pulse Vertical Second CVP.
143	TCON_OE1	B	Gate Driver/Row Output Enable
139	TCON_OE2	B	Gate Driver/Row Output Enable Second OE.

Table 21: SupraHD® 748 QFP Signals – LVDS & TCON Output

Pin #	Signal / Function	I/O Type	Description
144	TCON_STV1	B	Start Pulse Vertical Specifies the location of the first line during loading.
142	TCON_STV2	B	Start Pulse Vertical Second STV.
145	TCON_GPO	B	General Purpose Output Programmable by registers.

Table 22: SupraHD® 748 QFP Signals – Audio Input/Output

Pin #	Signal / Function	I/O Type	Description
211	AC_L1	AI	Audio Line-level Input #1 - Left
195	AC_R1	AI	Audio Line-level Input #1 - Right
210	AC_L2	AI	Audio Line-level Input #2 - Left
196	AC_R2	AI	Audio Line-level Input #2 - Right
209	AC_L3	AI	Audio Line-level Input #3 - Left
197	AC_R3	AI	Audio Line-level Input #3 - Right
208	AC_L4	AI	Audio Line-level Input #4 - Left
198	AC_R4	AI	Audio Line-level Input #4 - Right
249	DDX_LP	B	Audio DDX Left Channel Differential Output – Positive The ADATAIO[0] left-channel data is output on this line.
250	DDX_LN	B	Audio DDX Left Channel Differential Output – Negative The ADATAIO[0] left-channel data is output on this line.
246	DDX_RP	B	Audio DDX Right Channel Differential Output – Positive The ADATAIO[0] right-channel data is output on this line.
247	DDX_RN	B	Audio DDX Right Channel Differential Output – Negative The ADATAIO[0] right-channel data is output on this line.
251	DDX_LO_L	B	Audio DDX Left Channel Line-level Output The ADATAIO[1] left-channel data is output on this line.
252	DDX_LO_R	B	Audio DDX Right Channel Line-Level Output The ADATAIO[1] right-channel data is output on this line.
(*) 246	DDX_1a	B	Audio DDX Single-Ended Channel Output – ADATAIO[0] Left The ADATAIO[0] left-channel data is output on this line.
(*) 247	DDX_2a	B	Audio DDX Single-Ended Channel Output – ADATAIO[0] Right The ADATAIO[0] right-channel data is output on this line.
(*) 249	DDX_3a	B	Audio DDX Single-Ended Channel Output – ADATAIO[1] Left The ADATAIO[1] left-channel data is output on this line.
(*) 250	DDX_4a	B	Audio DDX Single-Ended Channel Output – ADATAIO[1] Right The ADATAIO[1] right-channel data is output on this line.
(*) 251	DDX_5a	B	Audio DDX Single-Ended Channel Output – ADATAIO[2] Left The ADATAIO[2] left-channel data is output on this line.
(*) 252	DDX_6a	B	Audio DDX Single-Ended Channel Output – ADATAIO[2] Right The ADATAIO[2] right-channel data is output on this line.
186	IEC958O	O	IEC958 (S/PDIF) Format Out This output carries audio data formatted in IEC 958 format. This signal can be used to drive external hardware audio decoders.
(*) 247	ACLK	B	Audio Clock This clock is 256 or 384 times the audio sample clock (LRCLK). It is used to generate the IEC-958 formatted output. It is divided to create the BCLK and LRCLK.

Table 22: SupraHD® 748 QFP Signals – Audio Input/Output

Pin #	Signal / Function	I/O Type	Description
(*) 249, or (*) 41	BCLK	B	Bit Clock Software determines whether this pin is an input or an output. As an output, it provides a serial audio clock with software selectable frequency. The clock is typically n times the audio sample rate, where n = 32, 48, 64, or 128. As an input, it represents a serial clock for LRCLK and ADATAO. It is nominally 64 times the audio sample rate. This pin is configured as an input upon device reset.
(*) 246, or (*) 40	LRCLK	B	Left/Right Channel Selector Software determines whether this pin is an input or an output. LRCLK indicates left and right channel selection on the serial audio input; it also indicates the audio sample rate. The polarity of LRCLK is programmable. As an output, LRCLK transitions at the sample rate. As an input, LRCLK determines the sample rate. In this case, the timing accuracy of LRCLK determines the positional accuracy of the output samples.
(*) 250	ADATAIO[0]	B	Serial Audio Data Output I/O Line 0 This is data line 0 of the I2S audio interface. Data can be either output or input (programmable), clocked by BCLK. All audio channels must have the same sample rate.
(*) 252	ADATAIO[1]	B	Serial Audio Data Output I/O Line 1 This is data line 1 of the I2S audio interface. Data can be either output or input (programmable), clocked by BCLK. All audio channels must have the same sample rate.
(*) 251	ADATAIO[2]	B	Serial Audio Data Output I/O Line 2 This is data line 2 of the I2S audio interface. Data can be either output or input (programmable), clocked by BCLK. All audio channels must have the same sample rate.
204	AC_VREFP	AO	Reference Voltage for Audio ADC - Positive Internally generated Audio ADC reference voltage pin for off-chip decoupling. Recommended: 10uF cap to AC_VREFN.
205	AC_VREFN	AO	Reference Voltage for Audio ADC - Negative Internally generated Audio ADC reference voltage pin for off-chip decoupling. Recommended: 10uF cap to AC_VREFP.
203	AC_RES	AO	Audio ADC External Resistor Reference Recommended: 9.53K (1%) resistor to GND.
207	AC_LCAP_P	AO	Audio ADC Left Channel Power Filter - Positive Recommended: 220pF cap to AC_LCAP_N.
206	AC_LCAP_N	AO	Audio ADC Left Channel Power Filter - Negative Recommended: 220pF cap to AC_LCAP_P.
199	AC_RCAP_P	AO	Audio ADC Right Channel Power Filter - Positive Recommended: 220pF cap to AC_RCAP_N.
200	AC_RCAP_N	AO	Audio ADC Right Channel Power Filter - Negative Recommended: 220pF cap to AC_RCAP_P.

Table 23: SupraHD® 748 QFP Signals – USB

Pin #	Signal / Function	I/O Type	Description
132	USB_DP	AI/O	USB 2.0 Data Line - Positive
130	USB_DN	AI/O	USB 2.0 Data Line - Negative
133	USB_REXT	AI	USB 2.0 External RSET Signal Input Recommended: 6.04K (1%) pull-down resistor.

Table 24: SupraHD® 748 QFP Signals – DDR2

Pin #	Signal / Function	I/O Type	Description
73	S0ADR12	O	DDR2 Address Line 12
124	S0ADR11	O	DDR2 Address Line 11
79	S0ADR10	O	DDR2 Address Line 10
72	S0ADR9	O	DDR2 Address Line 9
123	S0ADR8	O	DDR2 Address Line 8
74	S0ADR7	O	DDR2 Address Line 7
121	S0ADR6	O	DDR2 Address Line 6
76	S0ADR5	O	DDR2 Address Line 5
120	S0ADR4	O	DDR2 Address Line 4
77	S0ADR3	O	DDR2 Address Line 3
119	S0ADR2	O	DDR2 Address Line 2
78	S0ADR1	O	DDR2 Address Line 1
118	S0ADR0	O	DDR2 Address Line 0
106	S0DAT15	B	DDR2 Data Line 15
94	S0DAT14	B	DDR2 Data Line 14
108	S0DAT13	B	DDR2 Data Line 13
91	S0DAT12	B	DDR2 Data Line 12
92	S0DAT11	B	DDR2 Data Line 11
107	S0DAT10	B	DDR2 Data Line 10
95	S0DAT9	B	DDR2 Data Line 9
105	S0DAT8	B	DDR2 Data Line 8
102	S0DAT7	B	DDR2 Data Line 7
90	S0DAT6	B	DDR2 Data Line 6
98	S0DAT5	B	DDR2 Data Line 5
87	S0DAT4	B	DDR2 Data Line 4
88	S0DAT3	B	DDR2 Data Line 3
99	S0DAT2	B	DDR2 Data Line 2
89	S0DAT1	B	DDR2 Data Line 1
100	S0DAT0	B	DDR2 Data Line 0
109	S0DQS0P	B	DDR2 Data Strobe LDQS
110	S0DQS0N	B	DDR2 Data Strobe LDQSN
103	S0DQS1P	B	DDR2 Data Strobe UDQS
104	S0DQS1N	B	DDR2 Data Strobe UDQSN
97	S0DQM0	O	DDR2 Data Mask LDM
96	S0DQM1	O	DDR2 Data Mask UDM
83	S0BS0	O	DDR2 Bank Select 0
80	S0BS1	O	DDR2 Bank Select 1
82	S0BS2	O	DDR2 Bank Select 2
116	S0RASN	O	DDR2 Row Address Strobe
117	S0CASN	O	DDR2 Column Address Strobe
85	S0WEN	O	DDR2 Write Enable
112	S0CLKP	O	DDR2 Clock - Positive
113	S0CLKN	B	DDR2 Clock - Negative

Table 24: SupraHD® 748 QFP Signals – DDR2

Pin #	Signal / Function	I/O Type	Description
84	S0CKE	O	DDR2 Clock Enable
115	S0ODT	O	On Die Termination

Table 25: SupraHD® 748 QFP Signals – SPI

Pin #	Signal / Function	I/O Type	Description
191	SPI_CLK	O	SPI Clock
188	SPI_DI	I	SPI Master Data Input
187	SPI_DO	B	SPI Master Data Output
189	SPI_HOLD	B	SPI Master Hold
190	SPI_SELO	O	SPI Device Select 0

Table 26: SupraHD® 748 QFP Signals – UART & I2C

Pin #	Signal / Function	I/O Type	Description
9	UART0TX	B (open drain)	UART 0 Transmit
8	UART0RX	B (open drain)	UART 0 Receive
(*) 9	TVU_UART_TXD	B (open drain)	TVuC UART Transmit
(*) 8	TVU_UART_RXD	B (open drain)	TVuC UART Receive
38	I2C0C	B (open drain)	I2C Compatible Clock 0 This signal is the clock of the internal I2C interface between the TVuC and MIPS CPU. This signal can be externally monitored. It is not recommended to use this signal for other than monitoring.
39	I2C0D	B (open drain)	I2C Compatible Data 0 This signal is the data of the internal I2C interface between the TVuC and MIPS CPU. This signal can be externally monitored. It is not recommended to use this signal for other than monitoring.
193	I2C1C	B (open drain)	I2C Compatible Clock 1 This signal is the clock of the I2C interface to an external tuner. If the I2C interface to a tuner is not used, it is not recommended to use this signal.
194	I2C1D	B	I2C Compatible Data 1 This signal is the data of the I2C interface to an external tuner. If the I2C interface to a tuner is not used, it is not recommended to use this signal.

Table 27: SupraHD® 748 QFP Signals – General Purpose I/O

Pin #	Signal / Function	I/O Type	Description
48	GPIO_P10	B	General Purpose I/O Line 10
47	GPIO_P9	B	General Purpose I/O Line 9
45	GPIO_P8	B	General Purpose I/O Line 8
44	GPIO_P7	B	General Purpose I/O Line 7
43	GPIO_P6	B	General Purpose I/O Line 6
41	GPIO_P1	B	General Purpose I/O Line 1
40	GPIO_P0	B	General Purpose I/O Line 0
(*) 134	GPIO_S[16]	O	Secondary General Purpose I/O Line 16
(*) 190	GPIO_S[15]	O	Secondary General Purpose I/O Line 15
(*) 191	GPIO_S[14]	O	Secondary General Purpose I/O Line 14
(*) 187	GPIO_S[13]	B	Secondary General Purpose I/O Line 13
(*) 189	GPIO_S[12]	B	Secondary General Purpose I/O Line 12
(*) 186	GPIO_S[10]	O	Secondary General Purpose I/O Line 10
(*) 144	GPIO_S[9]	B	Secondary General Purpose I/O Line 9
(*) 143	GPIO_S[8]	B	Secondary General Purpose I/O Line 8
(*) 188	GPIO_S[7]	I	Secondary General Purpose I/O Line 7
(*) 194	GPIO_S[5]	B	Secondary General Purpose I/O Line 5
(*) 193	GPIO_S[4]	B (open drain)	Secondary General Purpose I/O Line 4
(*) 21	GPIO_TV_P29	B	TVuC General Purpose I/O Line 29
(*) 13	GPIO_TV_P27	B	TVuC General Purpose I/O Line 27
(*) 12	GPIO_TV_P26	B	TVuC General Purpose I/O Line 26
(*) 11	GPIO_TV_P23	B	TVuC General Purpose I/O Line 23
(*) 15	GPIO_TV_P15	B	TVuC General Purpose I/O Line 15
(*) 10	GPIO_TV_P13	B	TVuC General Purpose I/O Line 13
(*) 14	GPIO_TV_P12	B	TVuC General Purpose I/O Line 12
26	GPIO_TV_P7	B	TVuC General Purpose I/O Line 7
25	GPIO_TV_P6	B	TVuC General Purpose I/O Line 6
24	GPIO_TV_P5	B	TVuC General Purpose I/O Line 5
23	GPIO_TV_P4	B	TVuC General Purpose I/O Line 4
30	GPIO_TV_P3	B	TVuC General Purpose I/O Line 3
29	GPIO_TV_P2	B	TVuC General Purpose I/O Line 2
28	GPIO_TV_P1	B	TVuC General Purpose I/O Line 1
27	GPIO_TV_P0	B	TVuC General Purpose I/O Line 0

Table 28: SupraHD® 748 QFP Signals – TVuC and Main CPU Signals

Pin #	Signal / Function	I/O Type	Description
35	RESET_N	I	Reset In Input signal used to reset the TVuC. The TVuC has reset control of the rest of the IC which it can reset as needed via software control.
20	IRR	I	Infrared Receive
7	TVCPU_PWM0	O	TVuC PWM
15	POWER_CTL1	B	Regulator On/Off Control This signal is used to turn regulators on/off during standby mode.
14	POWER_CTL2	B	Regulator On/Off Control This signal is used to turn regulators on/off during standby mode.
13	WDT_EN_N	B	Watchdog Timer Enable At boot time, this signal enables or disables the 8051 watchdog timer.
12	BOOT_OPT/LED	B	Boot From MIPS At boot time, this signal must be pulled low for the MIPS to provide boot code to the 8051. After the boot is completed, this signal can be reconfigured as: LED Driver (Front Panel) This signal is used to drive the LED on the front panel.
(*) 40	BOOT_0	B	Boot Source 0 With BOOT_1, selects the boot source: BOOT[1:0] = 00b, boot from SPI flash. BOOT[1:0] = 01b, boot from UART.
(*) 41	BOOT_1	B	Boot Source 1 With BOOT_0, selects the boot source: BOOT[1:0] = 00b, boot from SPI flash. BOOT[1:0] = 01b, boot from UART.
10	INT0_N	B	External Interrupt 0 Input used to control timers for pulse width measurement.
11	INT1_N	B	External Interrupt 1 Input used to control timers for pulse width measurement.
21	INT2	B	External Interrupt 2 Input used to control timers for pulse width measurement.
(*) 13	T2	B	Timer/Counter 2 External Input
1	ADC8_IN2	AI	ADC8 Input Signal This signal is input to the ADC8 MUX and is designed as a general purpose input for the use of the system designer. This signal must be active while the device is in standby mode.
2	ADC8_IN1	AI	ADC8 Input Signal This signal is input to the ADC8 MUX and is designed as a general purpose input for the use of the system designer. This signal must be active while the device is in standby mode.
3	ADC8_IN0	AI	ADC8 Input Signal This signal is input to the ADC8 MUX and is designed as a general purpose input for the use of the system designer. This signal must be active while the device is in standby mode.
5	ADC8_RBIAIS	AI	ADC8 BIAS Voltage This signal should be pulled to GND via a 10K resistor.

Table 29: SupraHD® 748 QFP Signals – Clocks

Pin #	Signal / Function	I/O Type	Description
253	CLKIN_25M	AI	External Crystal Input An external 25 MHz crystal is connected to CLKIN_25M & CLKOUT_25M to provide the system clock. Typically CLKOUT_25M is buffered and sent to CLKIN_25M, which provides the clock reference to the digital portion of the IC. Crystal Specification: 50ppm 0-70 degrees C
254	CLKOUT_25M	AO	External Crystal Output An external 25 MHz crystal is connected to CLKIN_25M & CLKOUT_25M to provide the system clock. Typically CLKOUT_25M is buffered and sent to CLKIN_25M, which provides the clock reference to the digital portion of the IC. Crystal Specification: 50ppm 0-70 degrees C

Table 30: SupraHD® 748 QFP Signals – Power and Ground

Pin #	Signal / Function	I/O Type	Description
37	CORE_VDD	P	Core Voltage Power Supply (1.1V)
51	CORE_VDD	P	Core Voltage Power Supply (1.1V)
69	CORE_VDD	P	Core Voltage Power Supply (1.1V)
75	CORE_VDD	P	Core Voltage Power Supply (1.1V)
86	CORE_VDD	P	Core Voltage Power Supply (1.1V)
114	CORE_VDD	P	Core Voltage Power Supply (1.1V)
125	CORE_VDD	P	Core Voltage Power Supply (1.1V)
129	CORE_VDD	P	Core Voltage Power Supply (1.1V)
180	CORE_VDD	P	Core Voltage Power Supply (1.1V)
213	CORE_VDD	P	Core Voltage Power Supply (1.1V)
256	CORE_VDD	P	Core Voltage Power Supply (1.1V)
4	COREVDD_STBY	P	Core Voltage Standby Power Supply (1.1V)
17	COREVDD_STBY	P	Core Voltage Standby Power Supply (1.1V)
71	MVDD	P	Memory Interface Power Supply (1.8V)
101	MVDD	P	Memory Interface Power Supply (1.8V)
128	MVDD	P	Memory Interface Power Supply (1.8V)
42	IOVDD	P	IO Voltage Power Supply (3.3V)
46	IOVDD	P	IO Voltage Power Supply (3.3V)
49	IOVDD	P	IO Voltage Power Supply (3.3V)
137	IOVDD	P	IO Voltage Power Supply (3.3V)
141	IOVDD	P	IO Voltage Power Supply (3.3V)
181	IOVDD	P	IO Voltage Power Supply (3.3V)
192	IOVDD	P	IO Voltage Power Supply (3.3V)
212	IOVDD	P	IO Voltage Power Supply (3.3V)
6	IOVDD_STBY	P	IO Voltage Standby Power Supply (3.3V)

Table 30: SupraHD® 748 QFP Signals – Power and Ground

Pin #	Signal / Function	I/O Type	Description
36	IOVDD_STBY	P	IO Voltage Standby Power Supply (3.3V)
81	CVDD_MSD	P	Memory Interface Power Supply (1.1V)
93	CVDD_MSD	P	Memory Interface Power Supply (1.1V)
111	CVDD_MSD	P	Memory Interface Power Supply (1.1V)
122	CVDD_MSD	P	Memory Interface Power Supply (1.1V)
53	HDMI_VDD3P3	P	HDMI Interface Power Supply (3.3V)
64	HDMI_VDD3P3	P	HDMI Interface Power Supply (3.3V)
66	HDMI_VDD3P3	P	HDMI Interface Power Supply (3.3V)
70	HDMI_VDD3P3	P	HDMI Interface Power Supply (3.3V)
54	HDMI_VDD1V	P	HDMI Interface Power Supply (1.1V)
55	HDMI_VDD1V	P	HDMI Interface Power Supply (1.1V)
67	HDMI_VDD1V	P	HDMI Interface Power Supply (1.1V)
68	HDMI_VDD1V	P	HDMI Interface Power Supply (1.1V)
146	LDI_VDD	P	LVDS Power Supply (3.3V)
162	LDI_VDD	P	LVDS Power Supply (3.3V)
179	LDI_VDD	P	LVDS Power Supply (3.3V)
147	LDI_CVDD	P	LVDS TX Power Supply (1.1V)
178	LDI_CVDD	P	LVDS TX Power Supply (1.1V)
131	USB2_VDDA33T	P	USB 2.0 PHY Power Supply (3.3V)
127	DDRPLL_AVDD	P	DDR2 PLL Power Supply (3.3V) Connect to 3.3V through a 1.5Kohm resistor.
126	DDRPLL_GND	P	DDR2 PLL Ground Recommended: 10nF cap to DDRPLL_AVDD. Do not connect to GND on PCB.
243	VDDIO_SYSPLL	P	On-Chip PLL Power Supply (3.3V) Connect to 3.3V through a 1.5Kohm resistor.
255	CLK25M_VDD	P	Crystal Circuit Power Supply (3.3V)
201	AC_AVDD	P	Audio ADC Power Supply (3.3V)
202	AC_AGNDREF	P	Audio ADC Reference Ground
217	AFE_VPWR33	P	Analog Front End Power Supply (3.3V)
224	AFE_VPWR33	P	Analog Front End Power Supply (3.3V)
231	AFE_VPWR33	P	Analog Front End Power Supply (3.3V)
235	AFE_VPWR33_REF	P	AFE REF Power Supply (3.3V)
219	AFE_VPWR10	P	Analog Front End Power Supply (1.1V)
226	AFE_VPWR10	P	Analog Front End Power Supply (1.1V)
233	AFE_VPWR10	P	Analog Front End Power Supply (1.1V)
218	AFE_GND	P	Analog Front End Ground
225	AFE_GND	P	Analog Front End Ground
232	AFE_GND	P	Analog Front End Ground
236	AFE_VPWR33_PLL	P	AFE PLL Power Supply (3.3V)
240	AFE_VPWR10_PLL	P	AFE PLL Power Supply (1.1V)
237	AFE_PLL_GND	P	AFE PLL Ground
248	DDX_VDDIO	P	DDX Audio Power Supply (3.3V)
244	AUDPLL_VDDIO	P	Audio PLL Power Supply (3.3V)
245	AUDPLL_GND	P	Audio PLL Ground



SupraHD®-748-DTS-GEN-v1.1

Table 30: SupraHD® 748 QFP Signals – Power and Ground

Pin #	Signal / Function	I/O Type	Description
52	GND	P	Ground
e-pad	GND	P	Ground

Zoran Confidential
For LGD Only

2.3. Differences Between BGA and QFP Packages

The QFP package does not include all of the signals that the BGA package provides. The following table lists the signals available in the BGA package that are not available in the QFP package, including secondary, tertiary and fourth functions per ball. Note that the QFP package may have some more pins dedicated to certain power and ground rails than the BGA package allocates in balls, but that is only to compensate for the lack of a substrate in the QFP. There are no signals unique to the QFP package that are absent in the BGA package. Signals are listed according to function.

Table 31: SupraHD® 748 BGA Signals Absent from QFP Package

Signal	Description
AFE_SIF	SIF (Sound IF) Input
AFE_CVBS2	Composite Video Input 2
AFE_Y2	2nd Component Y Input
AFE_PB2	2nd Component Pb Input
AFE_PR2	2nd Component Pr Input
HDMI0_CP	HDMI0 Clock - Positive
HDMI0_CN	HDMI0 Clock - Negative
HDMI0_D2P	HDMI0 Data2 - Positive
HDMI0_D2N	HDMI0 Data2 - Negative
HDMI0_D1P	HDMI0 Data1 - Positive
HDMI0_D1N	HDMI0 Data1 - Negative
HDMI0_D0P	HDMI0 Data0 - Positive
HDMI0_D0N	HDMI0 Data0 - Negative
HDMI2_CP	HDMI2 Clock - Positive
HDMI2_CN	HDMI2 Clock - Negative
HDMI2_D2P	HDMI2 Data2 - Positive
HDMI2_D2N	HDMI2 Data2 - Negative
HDMI2_D1P	HDMI2 Data1 - Positive
HDMI2_D1N	HDMI2 Data1 - Negative
HDMI2_D0P	HDMI2 Data0 - Positive
HDMI2_D0N	HDMI2 Data0 - Negative
HDMI_ATEST	HDMI RX Analog Test.
LVDS_ATEST	LVDS TX Analog Test
PWM2	2nd Pulse Width Modulation Output
TCON_STH1	Start Pulse Horizontal
TCON_STH2	Start Pulse Horizontal for Send Half Line
AC_L5	Audio Line-level Input #5 - Left
AC_R5	Audio Line-level Input #5 - Right
AC_VCM	Audio ADC Common Mode Voltage
USB_ATEST	USB Analog Test
RDRIVER	RDRIVER Input (Driver Resistor Connection)
RDRIVER50	RDRIVER50 Input (Driver Resistor Connection)
RTERM	RTERM (Termination Resistor Connection)
S0VREF	DDR2 Voltage Reference
SPI_SEL1	SPI Device Select 1

Table 31: SupraHD® 748 BGA Signals Absent from QFP Package

Signal	Description
UART1TX	UART 1 Transmit
UART1RX	UART 1 Receive
GPIO_P16	General Purpose I/O Line 16
GPIO_P14	General Purpose I/O Line 14
GPIO_P12	General Purpose I/O Line 12
GPIO_P11	General Purpose I/O Line 11
GPIO_P5	General Purpose I/O Line 5
GPIO_P4	General Purpose I/O Line 4
GPIO_S[11]	Secondary General Purpose I/O Line 11
GPIO_S[3]	Secondary General Purpose I/O Line 3
GPIO_S[2]	Secondary General Purpose I/O Line 2
GPIO_TV_P24	TVuC General Purpose I/O Line 24
GPIO_TV_P21	TVuC General Purpose I/O Line 21
T0	Timer/Counter 0 External Input
T1	Timer/Counter 1 External Input
ADC8_IN6	ADC8 Input Signal
ADC8_IN3	ADC8 Input Signal
CLKIN_RTC	Real-Time Clock (RTC) External Crystal (Optional)
CLKOUT_RTC	Real-Time Clock (RTC) External Crystal (Optional)
CVSS_MSD	Memory Interface Ground (for 1.1V Supply)
CVSS_MSD	Memory Interface Ground (for 1.1V Supply)
CVSS_MSD	Memory Interface Ground (for 1.1V Supply)
CVSS_MSD	Memory Interface Ground (for 1.1V Supply)
GNDSYSPLL	On-Chip PLL Ground Recommended: 10nF cap to VDDIO_SYSPLL. Do not connect to GND on PCB.

Chapter 3. Operating Conditions

Please note that the information in this document is preliminary and subject to changes based upon the full characterization of the SupraHD® 748 device.

3.1. Heat Characterization

The operation of a chip causes the package temperature to rise. The chip junction temperature needs to be less than 125° C during chip operation. Generally, the thermal performance of the package is represented as follows:

$\Theta_{JA} = (T_j - T_a)/PD$, where Θ_{JA} = Thermal resistance from the junction ambient temperature, i.e., resistance to the heat flow from the chip to ambient air. T_j = junction temperature, T_a = ambient temperature, and PD = power dissipation.

3.2. Absolute Maximum Ratings

Table 32, "Absolute Maximum Ratings," on page 66 shows absolute maximum ratings.

Table 32: Absolute Maximum Ratings

Symbol	Parameter	Limit	Unit
TA	Ambient Operating Temperature	0 to 70	°C
TSTG	Storage Temperature	-55 to 150	°C
V_IN, V_OUT	Voltage on I/O Pins Relative to GND	3.63	V
VDD_33	Voltage on VDD Relative to GND (3.3V)	3.63	V
VDD_11	Voltage on VDD Relative to GND (1.1V)	1.21	V
VDD_18	Voltage on VDD Relative to GND (1.8V)	1.98	V

Stress above these limits may cause permanent damage to the device.

VDD is +/- 5%.

3.3. Power Dissipation

The table below shows power dissipation data for the SupraHD® 748. The values provided are for various process/voltage/temperature (PVT) conditions while the device is operating in the worst-case usage model from a power-consumption standpoint (Input mode: ATSC tuner). Heat shielding is provided by a perforated EMC shield/heat sink.

Table 33: Power Dissipation SupraHD® 748

PVT Conditions	Power (W)
Nominal process Nominal voltages 20C (room) ambient temperature	2.55
+3-sigma statistical FAST process Nominal voltages 70C ambient temperature	3.06
Nominal process +5% voltages 70C ambient temperature	2.92
+3-sigma statistical FAST process +5% voltages 70C ambient temperature	3.36

Junction temperature measured at 125° C, max top-of-package temperature = 103° C. Perforated EMC shield / heat sink is recommended to keep top-of-package temperature at or below maximum.

The table below shows a summary of typical and maximum current for each power supply. This is helpful for determining the required on-board power regulators. The current draw for each supply varies on the usage model so the worst-case usage model from a current draw standpoint is provided for each supply. These worst-case usage models are characterized by the input mode (e.g. input = ATSC tuner).

Table 34: Maximum Current Dissipation Per Supply: SupraHD® 748

Voltage	Typical Current (A)	Maximum Current (A)	Input Setting
Core digital 1.1V	1.190	1.710	ATSC Tuner
Core analog 1.1V	0.048	0.052	S-Video
CORE_VDDSTBY 1.1V	0.083	0.090	ATSC Tuner
MVDD 1.8V	0.369	0.407	HDMI 1080p
I/O 3.3V	0.200	0.270	Component YPbPr
IOVDD_STBY 3.3V	0.082	0.086	ATSC Tuner

MVDD 1.8V includes current draw of external DDR2.

The table below shows the standby power dissipation for the SupraHD® 748. The standby power is indicated for a 25MHz crystal or RC circuit (5MHz) as the clock source to the TVuC. The values are for the worst-case: +5% supplies, +3-sigma process and 70°C ambient.

Table 35: Standby Power Dissipation SupraHD® 748

Voltage	Value (V)	Max. Current (mA)	Power (mW)
XTAL = 25MHz			
COREVDD_STBY	1.155	8	8.0
IOVDD_STBY	3.465	10	33.0
Total	-	18	41.0
RC = 5MHz			
COREVDD_STBY	1.155	4	4.62
IOVDD_STBY	3.465	3	10.40
Total	-	7	15.02

3.4. Recommended DC Operating Conditions

Table 36, “Recommended DC Operating Conditions,” on page 68 shows recommended DC operating conditions per power supply.

Table 36: Recommended DC Operating Conditions

Symbol	Parameter	Min.	Typ.	Max.
IOVDD	I/O Supply Voltage	3.17 V	3.3 V	3.47 V
IOVDD_STBY	I/O TVuC/Standby Supply Voltage	3.17 V	3.3 V	3.47 V
LDI_VDD	LVDS 3.3V Supply Voltage	3.17 V	3.3 V	3.47 V
HDMI_VDD3P3	HDMI 3.3V Supply Voltage	3.17 V	3.3 V	3.47 V
USB2_VDDA33T	USB2.0 Supply Voltage	3.17 V	3.3 V	3.47 V
DDRPLL_AVDD	DDR2 PLL 3.3V Supply Voltage	3.17 V	3.3 V	3.47 V
VDDIO_SYSPLL	System PLL Supply Voltage	3.17 V	3.3 V	3.47 V
CLK25M_VDD	Clock Circuit Supply Voltage	3.17 V	3.3 V	3.47 V
AC_AVDD	Audio ADC Supply Voltage	3.17 V	3.3 V	3.47 V
AFE_VPWR33	AFE 3.3V Supply Voltage	3.17 V	3.3 V	3.47 V
AFE_VPWR33_REF	AFE Reference Voltage	3.17 V	3.3 V	3.47 V
AFE_VPWR33_PLL	AFE PLL 3.3V Supply Voltage	3.17 V	3.3 V	3.47 V
DDX_VDDIO	DDX Audio Supply Voltage	3.17 V	3.3 V	3.47 V
AUDPLL_VDDIO	Audio PLL Supply Voltage	3.17 V	3.3 V	3.47 V
MVDD	DDR2 Supply Voltage	1.71 V	1.8 V	1.89 V
CORE_VDD	Core Supply Voltage	1.05 V	1.1 V	1.16 V
LDI_CVDD	LVDS 1.1V Supply Voltage	1.05 V	1.1 V	1.16 V
HDMI_VDD1V	HDMI 1.1V Supply Voltage	1.05 V	1.1 V	1.16 V

Table 36: Recommended DC Operating Conditions

Symbol	Parameter	Min.	Typ.	Max.
CVDD_MSD	DDR2 MSD Supply Voltage	1.05 V	1.1 V	1.16 V
COREVDD_STBY	Core TVuC/Standby Supply Voltage	1.05 V	1.1 V	1.16 V
AFE_VPWR10	AFE 1.1V Supply Voltage	1.05 V	1.1 V	1.16 V
AFE_VPWR10_PLL	AFE PLL 1.1V Supply Voltage	1.05 V	1.1 V	1.16 V
V_IH	Pad input voltage high	2.0 V	-	-
V_IL	Pad input voltage low	-	-	0.8 V
V_OH	Pad output voltage high	2.4 V	-	-
V_OL	Pad output voltage low	-	-	0.4 V
I_IN	Input Leakage Current	-10 uA	-	10 uA
C_IN	Input Capacitance	-	8.0 pF	-
C_OUT	Output Capacitance	-	8.0 pF	-

3.5. DDR2 Electrical Characteristics

Table 37, “DDR2 Interface Absolute Maximum Ratings,” on page 69 shows the DDR2 interface absolute maximum ratings, Table 38, “DDR2 DC Electrical Specifications,” on page 70 shows the DDR2 DC electrical specifications and Table 39, “DDR2 AC Electrical Specifications,” on page 70 shows the DDR2 AC electrical specifications. The SupraHD® 748 memory interface is implemented with this interface.

Table 37: DDR2 Interface Absolute Maximum Ratings

Symbol	Parameter	Conditions	Min	Typ	Max
MVDD	SSTL_18 post-driver power SSTL_18 level shifter power		1.71 V	1.8 V	1.89 V
CORE_VDD	SSTL_18 pre-driver and core logic power		1.05 V	1.1 V	1.16 V
S0VREF	SSTL_18 reference voltage supply	MVDD / 2	0.81 V	0.9 V	0.945 V
VI	Pad input voltage		0 V	MVDD	2.0 V
VO	Pad output voltage		0 V	MVDD	2.0 V
VIC	Core input voltage		0 V	CORE_VDD	1.16 V
VOC	Core output voltage		0 V	CORE_VDD	1.16 V

Table 38: DDR2 DC Electrical Specifications

Symbol	Parameter	Conditions	Min	Typ	Max
VDDQ	SSTL@ I/O voltage (MVDD)	+/- 5%	1.71 V	1.8 V	1.89 V
VREF	SSTL@ reference voltage (S0VREF)	$0.5 * V_DDQ$	$0.45 * VDDQ$	$0.50 * VDDQ$	$0.525 * VDDQ$
VTT	Termination voltage	-	$VREF - 0.04 V$	VREF	$VREF + 0.04 V$
VIH	DC input logic high	-	$VREF + 0.125 V$	-	$VDDQ + 0.3 V$
VIL	DC input logic low	-	-0.3 V	-	$VREF - 0.125 V$
IOH	Output minimum source DC current	$VOH = 1.68 V$	-15.2 mA	-	-
IOL	Output minimum sink DC current	$VOL = 0.54 V$	15.2 mA	-	-

Table 39: DDR2 AC Electrical Specifications

Symbol	Parameter	Conditions	Min	Typ	Max
V_IN	AC differential input voltage against VREF	$ VIN - VREF $	0.35 V		
VOUT1	AC voltage across RT	$IO_{min} = 15.2mA$	0.38 V		
VOUT2	AC voltage across RS+RT	$IO_{min} = 15.2mA$	0.76 V		
VIH	AC input logic high	$(VREF + VIN)$	$VREF + 0.35 V$	-	-
VIL	AC input logic low	$(VREF - VIN)$		-	$VREF - 0.35 V$
VOH	AC output logic high	$(VTT_{max} + VOUT2)$	$VTT_{max} + 0.57V$	-	-
VOL	AC output logic low	$(VTT_{max} - VOUT2)$	-	-	$VTT_{min} - 0.76 V$
VSWING	Input maximum swing	Peak-to-peak	-	-	1.5 V
SLEW	Input minimum slew rate	$VIH_{min} - VIL_{max} / t$	1.0 V/ns	-	-

3.6. Digital Video Electrical Characteristics

Table 40: Primary Digital Video I/O Electrical Specifications

Symbol	Parameter	Conditions	Min	Typ	Max
VDDQ	I/O voltage (IOVDD)	+/- 5%	3.135 V	3.3 V	3.465 V
VIH	DC input logic high		2.0 V	3.0 V	$VDDQ + 0.3V$
VIL	DC input logic low		-0.3 V	0 V	0.8 V
VOH	DC output high		2.4 V	3.0 V	$VDDQ + 0.3V$
VOL	DC output low		-0.3 V	0.1 V	0.4 V

Chapter 4. SupraHD® 748 Signal Maps, Packaging and Ordering Information

4.1. SupraHD® 748 Order Information

Please contact your local Zoran Sales Representative for information about ordering the SupraHD® 748.

fully-Green means lead and other heavy metals have been eliminated, in addition to the molding compound, die attachment, and BGA substrate material being halide-free.

The full part numbers for the BGA and QFP packages of the SupraHD® 748 integrated DTV SoCs are, respectively:

- ZR39748BGCG, fully-Green part
- ZR39748ELCG, fully-Green part

4.2. SupraHD® 748 Ball and Pin Maps

Figure 5, “ZR39748 BGA Ballout - Top View, Looking Down (Part 1 of 2)” on page 72 through Figure 7, “ZR39748 QFP Pins - Top View, Looking Down” on page 74 show the top view of the ZR39748 BGA and QFP packages with physical ball/pin assignments (for BGA, as though the material is transparent and you are viewing down through the top of the package). Since a BGA package is an array, there is no Ball 1. Instead, all balls are identified by the alpha-numeric locator shown around the edges of the figure, and there is a coordinate ball labeled “A1.” Pad A1 is in the upper right-hand corner of the drawing; this pin is identifiable on the package by a gold mark on the top corner of the package correlating to pin A1 beneath (see Figure 8).

	1	2	3	4	5	6	7	8	9	10	11
A	HDMI_CEC	IRR	VGA_SDA	IF_AGC	BOOT_OPT/ LED	UART0TX	CLKIN_RTC	CLKOUT_25M	DDX_LO_L	DDX_LP	DDX_RP
B	GPIO_TV_P1	INT2	VGA_SCL	T0	WDT_EN_N	T1	CLKOUT_RTC	CLKIN_25M	DDX_LN	DDX_RN	AFE_SOY
C	GPIO_TV_P3	GPIO_TV_P4	GPIO_TV_P5	GPIO_TV_P6	POWER_CTL2	INT0_N	UART0RX	ADC8_IN0	ADC8_IN2	DDX_LO_R	AFE_SOG
D	HDMI_SVSN	GPIO_TV_P2	GPIO_TV_P1	GPIO_TV_P0	POWER_CTL1	INT1_N	TVCPU_PWM0	ADC8_IN1	ADC8_IN3	AFE_FLTOUT2	AFE_VPWR3_PLL
E	RESET_N	HDMI0_HPD	HDMI0_SDA	HDMI0_SCL							
F	GPIO_P4	GPIO_P0	GPIO_P3	GPIO_P2		ADC8_RBIAS	CLK25M_VDD	DDX_VDDIO	AUDPLL_VDDIO	AFE_VPWR0_PLL	AFE_VPWR3_REF
G	GPIO_P8	GPIO_P7	GPIO_P6	GPIO_P5		IOVDD_STBY	COREVDD_STBY	VDDIO_SYSPLL			
H	GPIO_P12	GPIO_P11	GPIO_P10	GPIO_P9		IOVDD_STBY			AUDPLL_GND	GND	AFE_PLL_GND
J	HDMI2_CN	HDMI2_CP	AFE_HSYNC_IN	GPIO_P14		CORE_VDD			GND	GND	GND
K	HDMI2_D0N	HDMI2_D0P	GND						GND	GND	GND
L	HDMI2_D1N	HDMI2_D1P	GPIO_P1		IOVDD	IOVDD			GND	GND	GND
M	HDMI2_D2N	HDMI2_D2P	GND						GND	GND	GND
N	HDMI1_CN	HDMI1_CP	GPIO_P16		HDMI_VDD03P3	HDMI_VDD03P3			GND	GND	GND
P	HDMI1_D0N	HDMI1_D0P	GND						GND	GND	GND
R	HDMI1_D1N	HDMI1_D1P	GND		CORE_VDD	HDMI_VDD1V			HDMI_VDD03P3	GND	CVSS_MSD
T	HDMI1_D2N	HDMI1_D2P	GND		CORE_VDD	CORE_VDD					
U	HDMI0_CN	HDMI0_CP	HDMI0_ATEST		HDMI0_REX1				MVDD		CVDD_MSD
V	HDMI0_D0N	HDMI0_D0P	GND						MVDD	MVDD	CVDD_MSD
W	HDMI0_D1N	HDMI0_D1P	GND								
Y	HDMI0_D2N	HDMI0_D2P	GND	GND	GND	GND	GND	GND	GND	GND	GND
AA	S0ADR9	S0ADR5	S0ADR1	S0BS1	S0BS2	S0CKE	S0DAT4	S0DAT6	S0DAT14	S0DAT9	S0DQM0
AB	S0ADR12	S0ADR7	S0ADR3	S0ADR10	S0BS0	S0WEN	S0DAT3	S0DAT1	S0DAT12	S0DQM1	S0DAT11

Figure 5: ZR39748 BGA Ballout - Top View, Looking Down (Part 1 of 2)

12	13	14	15	16	17	18	19	20	21	22	
AFE_IPF	AFE_SF	AFE_CVBS2	AFE_SVID_Y	AFE_SVID_C	AC_L4	AC_RS	AC_R3	AC_R1	UCSC	UCSD	A
AFE_IPN	AFE_CVBS1	AFE_VGA_R	AFE_VGA_G	AFE_VGA_B	AC_L3	AC_L5	AC_R4	AC_R2	SPICLK	SPID	B
AFE_PICOUT	AFE_PR	AFE_Y	AFE_B	AC_L1	AC_L2	TMS	WCSB0	SP_H0ID	SP_SEL0	SP_OD	C
AFE_RST	ADC8_N6			AC_RCAP_N	AC_RCAP_P	TDO	TDI	TCCK	LVD_S_CUP	LVD_S_CUN	D
								UART1RX	LVD_S_D0UP	LVD_S_D0UN	E
AFE_V_PWR33	AFE_V_PWR10	AC_LCAP_P	AC_LCAP_N	AC_A_VDD	AC_RES	AC_VCM		UART1TX	LVD_S_D1UP	LVD_S_D1UN	F
					AC_VREFP	AC_VREFN		GND	LVD_S_D2UP	LVD_S_D2UN	G
AFE_GND	GND	GND	AC_AGNDRFP		CDRE_VDD				LVD_S_D3UP	LVD_S_D3UN	H
GND	GND	GND	GND						LVD_S_D4UP	LVD_S_D4UN	J
GND	GND	GND	GND		IOVDD	IOVDD			LVD_S_D5UP	LVD_S_D5UN	K
GND	GND	GND	LVD6_ATEST		LDLVDD	LDLVDD			LVD6_D0UP	LVD6_D0UN	L
GND	GND	GND	GND						LVD6_D1UP	LVD6_D1UN	M
GND	GND	GND	GND		LDLVDD	LVD6_REXT			LVD6_D2UP	LVD6_D2UN	N
GND	GND	GND	GND						LVD6_D3UP	LVD6_D3UN	P
CVSS_MSD	CVSS_MSD	GND	DPRLL_GND		CDRE_VDD	CDRE_VDD			LVD6_D4UP	LVD6_D4UN	R
					USB2_VDDA33T	USB2_REXT			TC0N_GPD	TC0N_STV1	T
	MVDD	MVDD		DDRPLL_AVDD	DDRPLVDD				TC0N_STV2	TC0N_STV3	U
	SDVREF	MVDD		RTEPM	DDRPLVDD			USB2_ATEST	TC0N_CPV1	TC0N_D02	V
								TC0N_RVS	TC0N_CPV2	USB2_DN	W
GND	GND	GND	GND	GND	GND	GND	GND	PWM	SP_SEL1	SDADR1	Y
SDAT2	SDAT7	SDQS1P	SDAT15	SDATA13	SDQSDP	SDCLKN	SDRASN	SDADR0	SDADR4	SDADR8	AA
SDAT5	SDAT9	SDATA10	SDQSLN	SDATA16	SDQSDN	SDCLNP	SDCDT	SDCA5N	SDADR2	SDADR6	AB

Figure 6: ZR39748 BGA Ballout - Top View, Looking Down (Part 2 of 2)

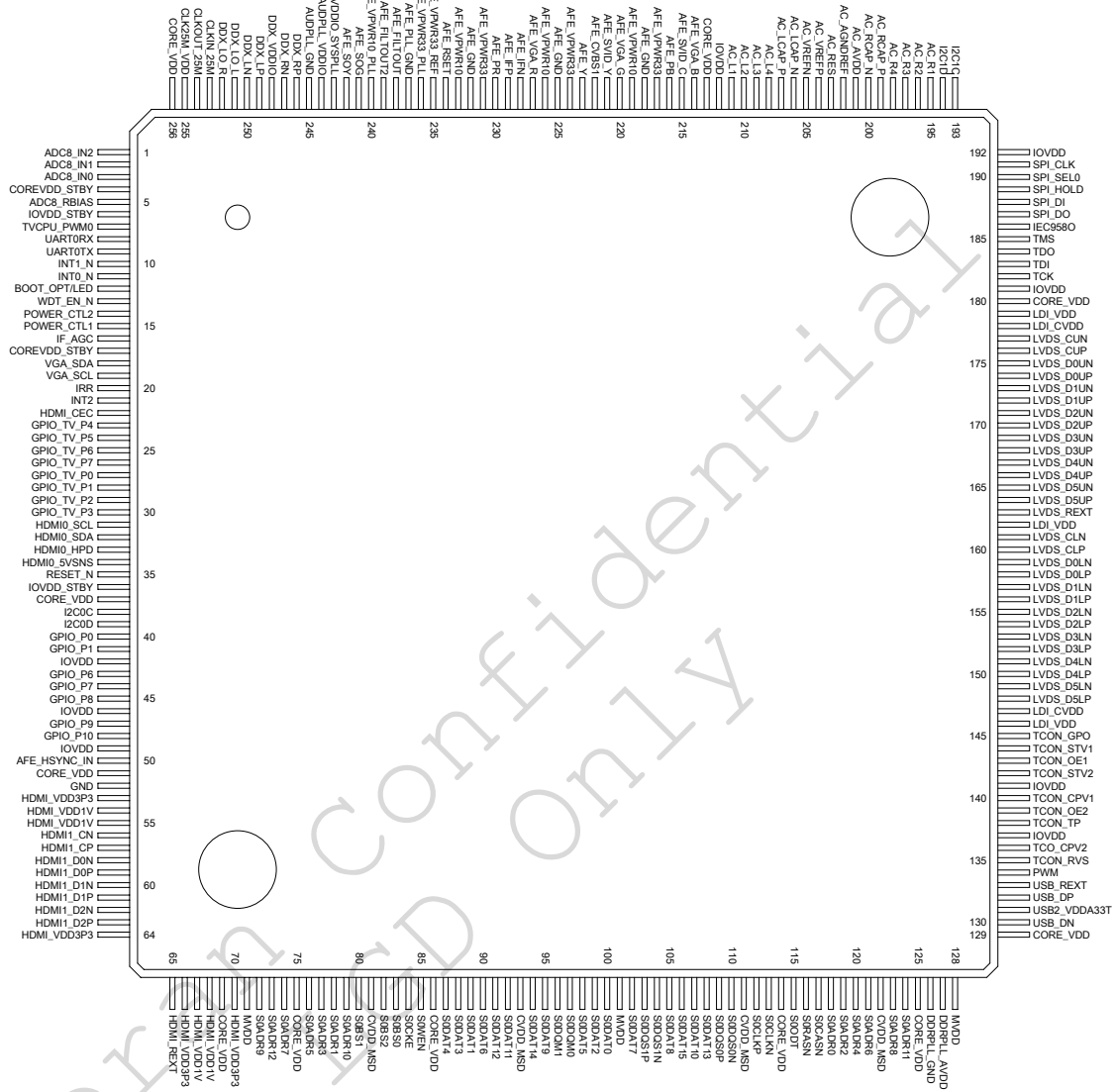


Figure 7: ZR39748 QFP Pins - Top View, Looking Down

4.3. BGA Package Mechanicals

All measurements are in millimeters unless otherwise noted.

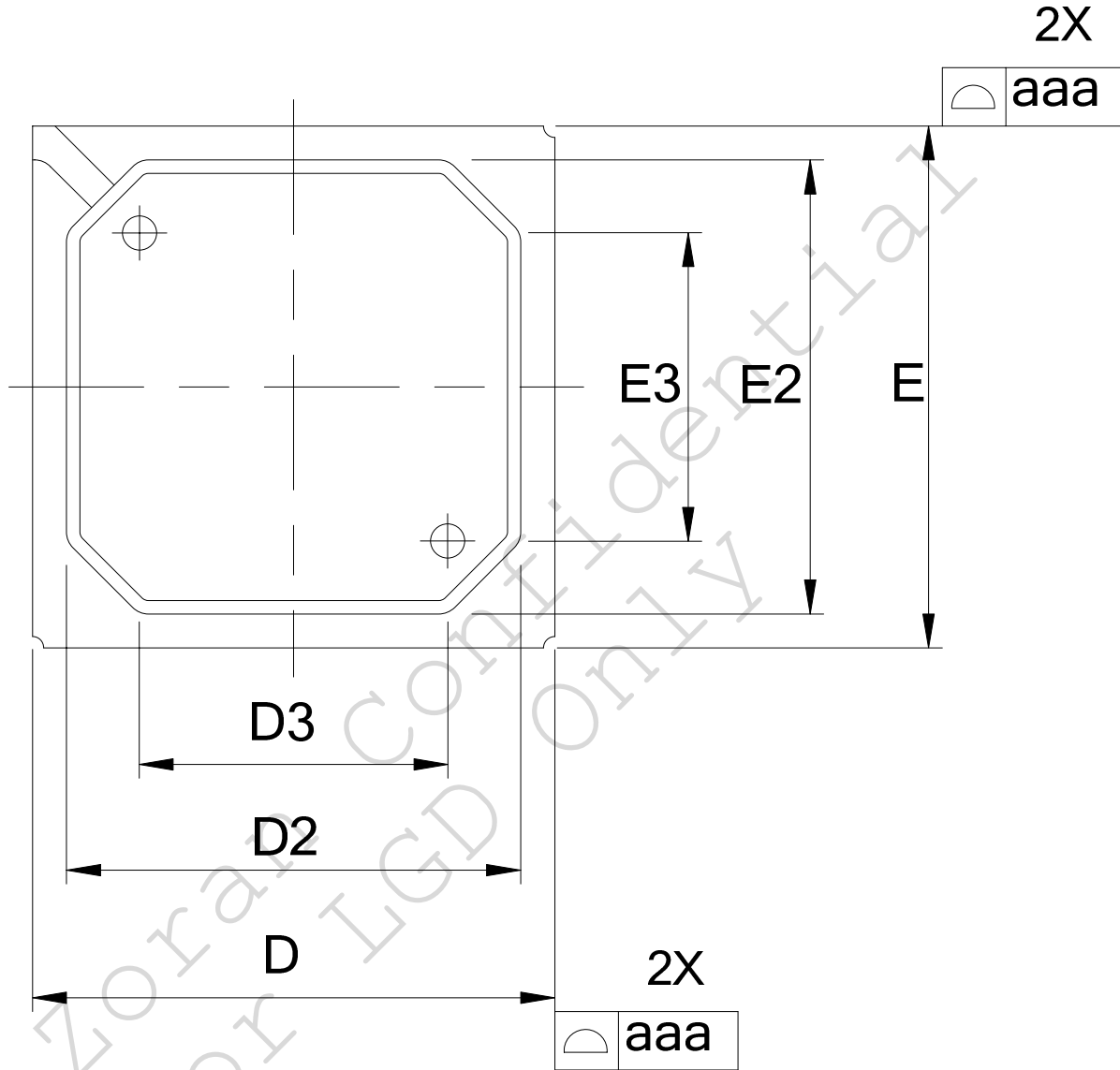


Figure 8: ZR39748 BGA Physical Package (Top View)

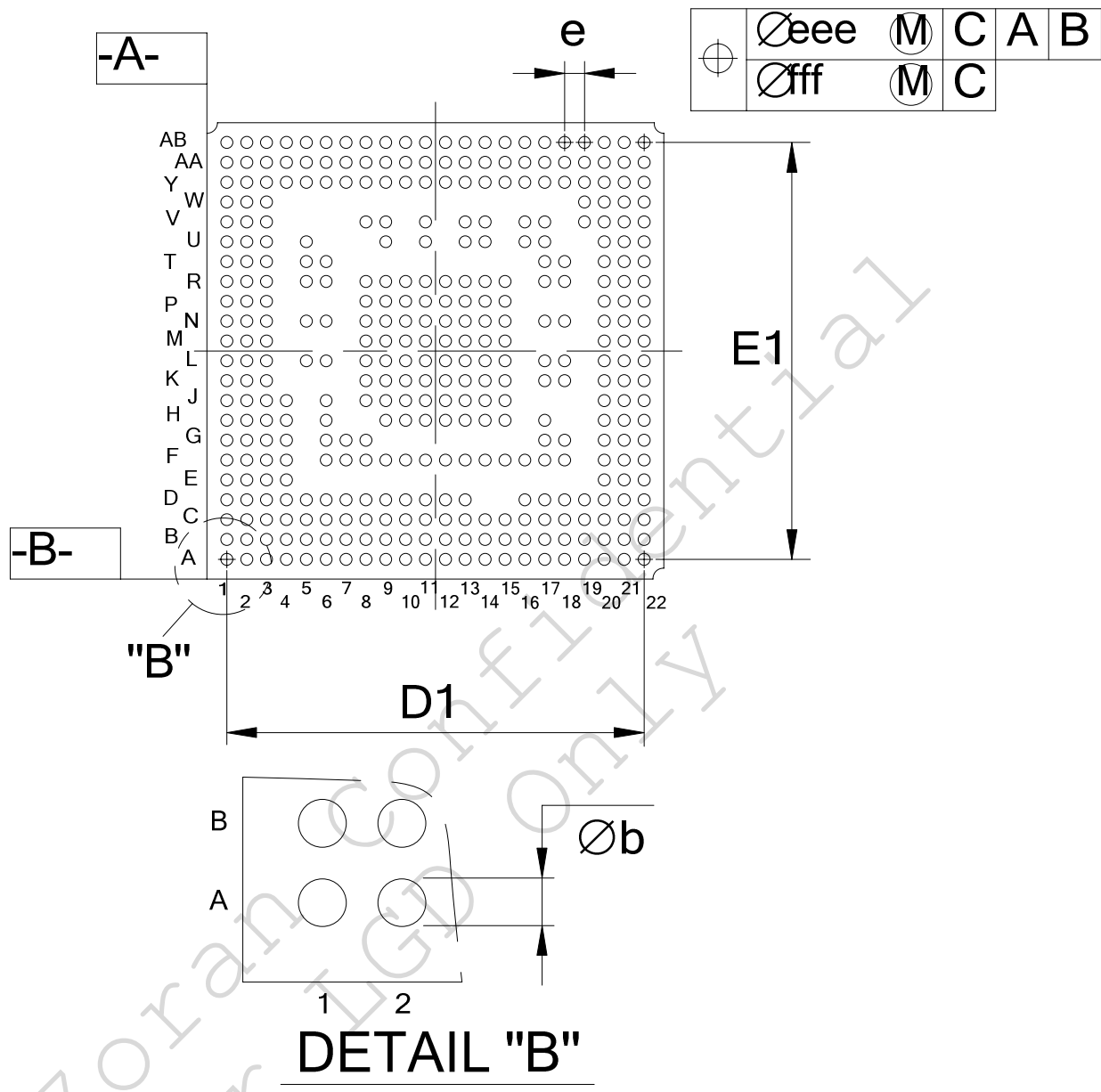


Figure 9: ZR39748 BGA Physical Package (Bottom View)

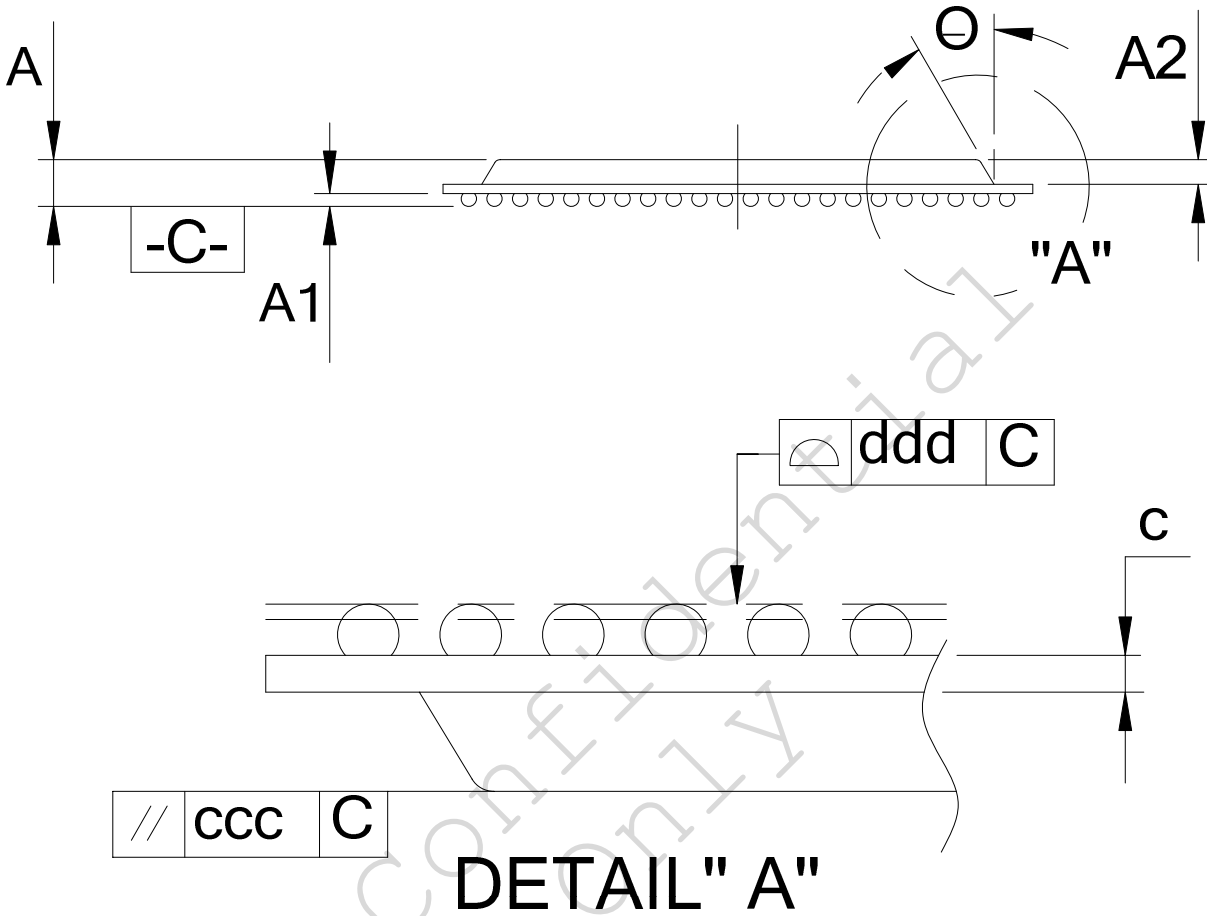


Figure 10: ZR39748 BGA Physical Package (Side View)

Table 41, "BGA Package Physical Parameters," on page 78 shows the physical measurement parameters of the package.

Table 41: BGA Package Physical Parameters

Name		Symbol	Common Dimensions
Package			PBGA
Body Size	X	D	23.00 +/- 0.200
	Y	E	23.00 +/- 0.200
Ball Pitch		e	1.000
Total Thickness		A	1.830 +/- 0.130
Mold Thickness		A2	0.970 +/- 0.050
Substrate Thickness		c	0.360 +/- 0.040
Ball Diameter		b	0.600 +/- 0.100
Stand Off		A1	0.500 +/- 0.100
Mold Area	X	D2	20.00 +/- 0.200
	Y	E2	20.00 +/- 0.200
Chamfer	X	D3	13.60
	Y	E3	13.60
Package Edge Tolerance		aaa	0.200
Mold Flatness		ccc	0.250
Coplanarity		ddd	0.150
Ball Offset (Package)		eee	0.250
Ball Offset (Ball)		fff	0.100
Ball Count		n	365
Mold Angle		θ	30 degrees TYP
Edge Ball Center to Center	X	D1	21.000
	Y	E1	21.000

4.4. QFP Package Mechanicals

All measurements are in millimeters unless otherwise noted.

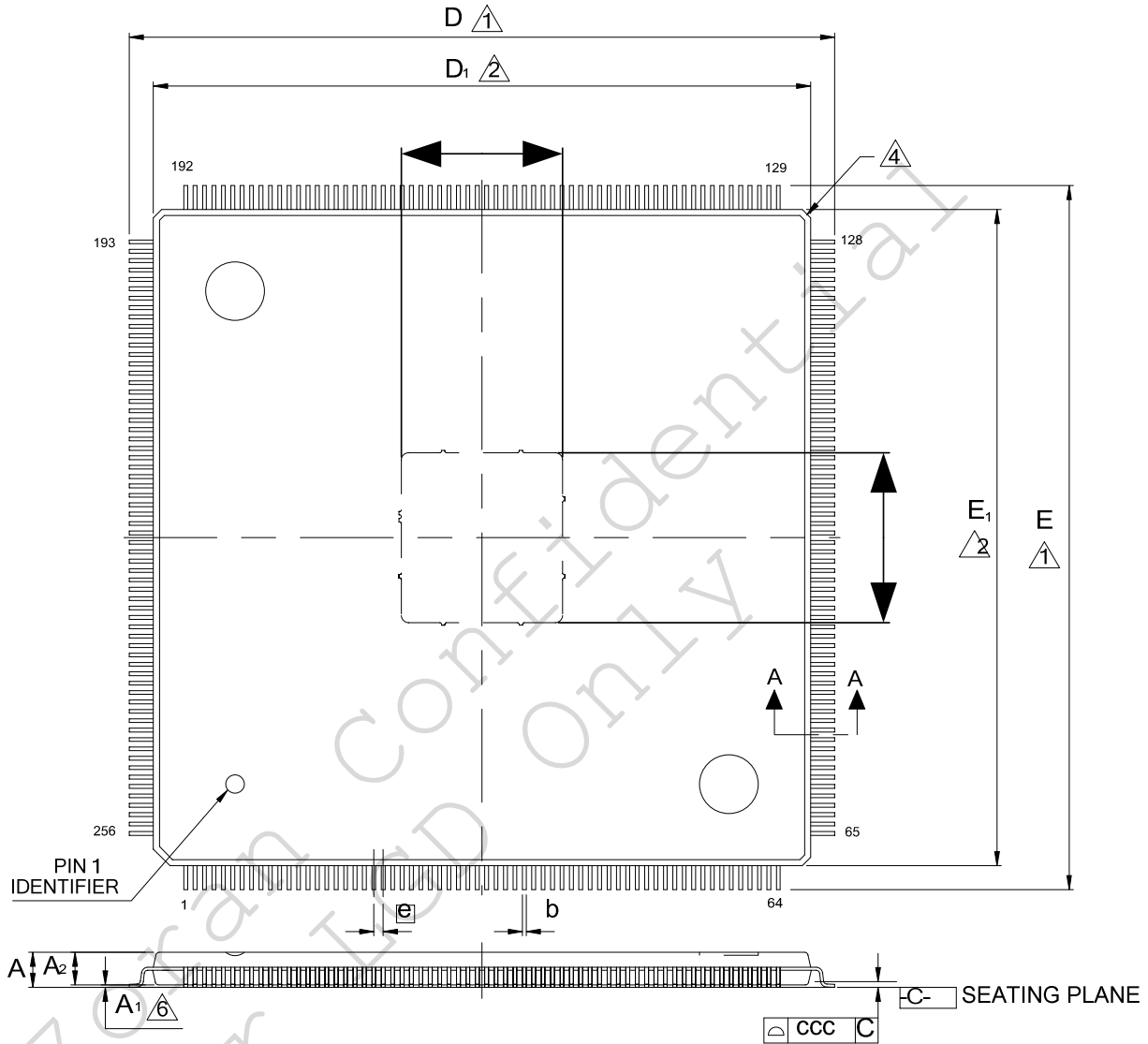


Figure 11: ZR39748 QFP Physical Package (Top & Side Views)

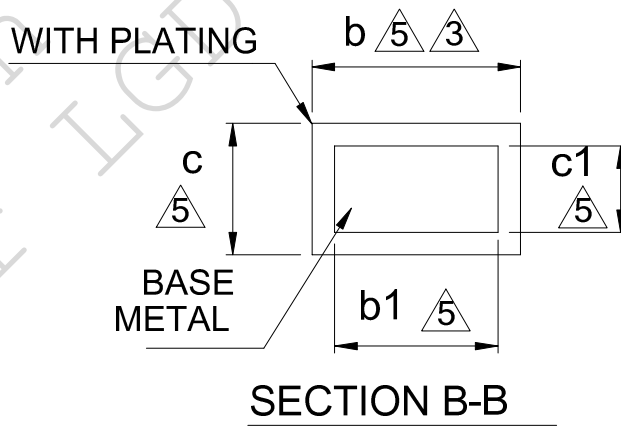
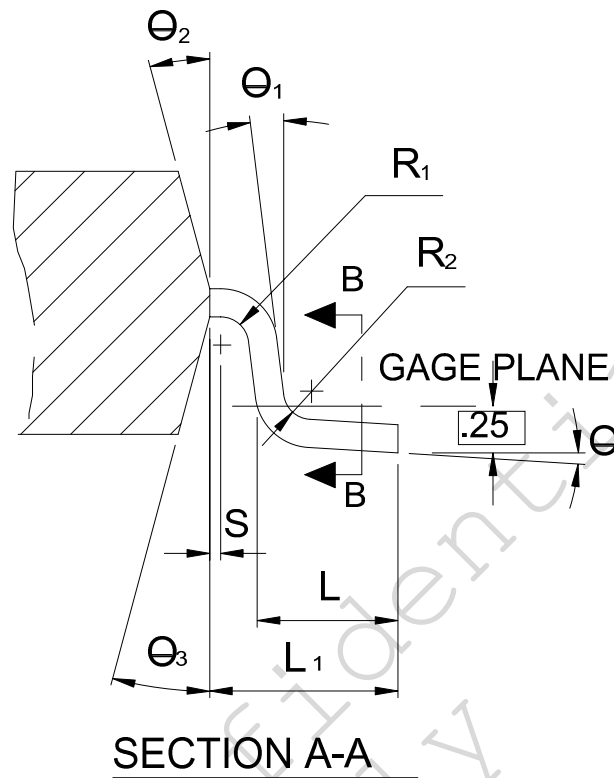


Figure 12: ZR39748 QFP Physical Package (Section Views)

Table 42, "QFP Package Physical Parameters," on page 81 shows the physical measurement parameters of the package along with notes (designated by numbers in triangles) for the above figures.

Table 42: QFP Package Physical Parameters

Symbol	Dimension Min.	Dimension Nom.	Dimension Max.
A	-	-	1.60
A1	0.05	-	-
A2	1.35	1.40	1.45
b	0.13	0.18	0.23
b1	0.13	0.16	0.19
c	0.12	-	0.20
c1	0.13 REF		
D	29.85	30.00	30.15
D1	27.90	28.00	28.10
E	29.95	30.00	30.15
E1	27.90	28.00	28.10
e	0.40 BSC		
L	0.45	0.60	0.75
L1	1.00 REF		
R1	0.15 REF		
R2	0.15 REF		
S	0.21 REF		
θ	0 degrees	3.5 degrees	7 degrees
θ1	7 degrees REF		
θ2	12 degrees REF		
θ3	12 degrees REF		
ccc	0.08		

Notes:

1. To be determined at seating plane –C–
2. Dimensions D1 and E1 do not include mold protrusion. D1 and E1 are maximum plastic body size dimensions including mold mismatch.
3. Dimension b does not include dambar protrusion. Dambar cannot be located on the lower radius or the foot.
4. Exact shape of each corner is optional.
5. These dimensions apply to the flat section of the lead between 0.10mm and 0.25mm from the lead tip.
6. A1 is defined as the distance from the seating plane to the lowest point of the package body.

Appendix A: Change History

Table 43: History of Document Changes

Revision Number	Changes
1.0	Initial document (chris 4/30/10)
1.1	- Added ADC8, removed 3rd I2C from Figure 1 "Basic SupraHD® 748 IC Block Diagram" on page 1 and Figure 3 "SupraHD® 748 Detailed Block Diagram" on page 12. - Added clarification for audio format support (I2S vs. others). - Corrected maximum DDR2 size supported, typical memory size usage. - Added additional details to graphics processing support. - Corrected UART baud rate from 110K to 115200. - Removed TVCPU_I2C1C & TVCPU_I2C1D functionality. - Removed FP_PWR_KEY functionality. - Switched HDMI0_HPD and HDMI0_SVSNS. - Switched HDMI1_HPD and HDMI1_5VSENSE. - Corrected ACLK and ADATAIO[2] signal placement. - Added single-ended DDX options. - Added T2 as additional function to WDT_EN_N signal. - Added I2S audio options to GPIO_P[.] - Changed BOOT_OPT to be low at 8051 boot. - Changed QFP pin 49 from GPIO_P11 to IOVDD. - Removed PWM2 from QFP signals. - Corrected standby currents typos in Table 34, "Maximum Current Dissipation Per Supply: SupraHD® 748" on page 67. - Removed ripple voltages from Table 36, "Recommended DC Operating Conditions" on page 68. - Minor formatting, spelling and grammar corrections throughout.



Appendix B: Document Improvement

The Zoran customer support team recognizes the need for accurate and easy to use documentation, however some document data may need clarification or correction. The first line of contact should be your Zoran field applications engineer.

When requesting document assistance and/or improvement, please provide the following identifying information:

- Document Title (cover sheet)
- Publication Date (cover sheet)
- Version Number (cover sheet)
- Document Number (inside cover sheet)
- Paragraph Number (which statement is in question?)
- Figure Number (which area of the figure is of concern?)
- Table Number (which row and column are of interest?)

If you have support questions or concerns, please e-mail us at:

DTVSupport@zoran.com